

CS301 Quiz No.01 Solution Fall2021

For a perfect binary tree of height 4, what will be the sum of highest of node

26

If Ahmed is cousin of Ali and Ali is cousin of Asad then Ahmed is also cousin of Asad. This statement has the following property

Transitivity

Suppose there are 100 elements in an equivalence class, so initially there will be 100 trees, the collection of these trees is called.

Forest

The percolate Down procedure will move the smaller value and bigger value.

Up, down

For a perfect binary tree of height h , having N nodes, the sum of height of nodes is

$N-h-1$

If we want to find median of 50 elements, then after applying builtHeap method, how

Manytime deleteMin method will be called?

25

The Expression

If(!heap-> isEmpty ())

Check

Heap is not empty

Which of the following heap method increase the value if Key at position 'p' by the Amount 'delta'?

Increase Key(p,delta)

Given the values are the array representation of heap; 12 23 26 31 34 44 56 64 78 100

What is the 5th smallest element in the given heap?

34

Which of the following data structure linear type

All of above

in a program a reference variable with name 'x' can be declared as

int &'x';

Object (object oriented by pointers) are called anonymous object

Friend

in the perspective each process divided into.....section

4

In tree the search operation is.....as compared to the linked list

Very fast

Level order transversal for binary tree search can be implemented.....

Only through non recursive method

.....is an area in computer memory that is used to allocate memory dynamically

Stock

Suppose we want to create the right child of 'p' and set the value x in it. which of the following statement is correct in this operation

Set Right (x,p)

In level order traversal for binary search tree of each level we visit node in.....order

Right to left

We can calculate the.....of a subtree counting the level from the bottom

Data items

Ais a tree in which every level except possibly the last is completely filled

Complete binary tree

Which of the following calling method does not change the original value of variable passed to a function as an argument?

Call by passing as value

The _____ of a binary tree is the maximum level of its leaves (also called the depth).

Select correct option:

Level

Width

Height

None of the above

Question # 2 of 10 (Start time: 06:40:41 PM) Total Marks: 1

If we write functions for recursive and non recursive inorder traversal method of BST, what will be the difference between its functions prototypes?

Select correct option:

Different return types

Different function names

Different arguments list

Nothing will be different

Question # 3 of 10 (Start time: 06:40:41 PM) Total Marks: 1

binary tree whose every node has either zero or two children is called _____

Select correct option:

Complete binary tree

Binary search tree

Strictly binary tree

None of above

Question # 4 of 10 (Start time: 06:40:41 PM) Total Marks: 1

Which one of the following calling method does not change the original value of the argument in the calling function?

Select correct option:

Call by passing reference of the argument

Call by passing the address of the argument

Call by passing the value of the argument

None of the given options

Question # 5 of 10 (Start time: 06:40:41 PM) Total Marks: 1

Longest path from root node to farthest leaf node is called _____ of tree

Select correct option:

Level

Length

Depth

Node level

Question # 6 of 10 (Start time: 06:40:41 PM) Total Marks: 1

Which of the following statement is false?

Select correct option:

Arrays are dense lists and static data structure

data elements in linked list need not be stored in adjacent space in memory

pointers store the next data element of a list

linked lists are collection of the nodes that contain information part and next pointer

The _____ of a node in a binary tree is defined as the height of its left subtree minus height of its right subtree.

Select correct option:

Height

Balance

Width

None of the above

A tree is an AVL tree if

Select correct option:

Any one node fulfills the AVL condition

At least half of the nodes fulfill the AVL condition

All the nodes fulfill the AVL condition

None of the given options

In a program a reference variable, say x, can be declared as

Select correct option:

int &x ;

int *x ;

int x ;

None of the given options

In which of the following functions, the value of variable cannot be changed in function body?

Select correct option:

int cube(int num)

int cube(int& num)

int cube(const int& num)

int cube(int* num)

What is the maximum depth of recursive calls a function may make?

n (where n is the argument)

Here is a small function definition:

```
void f(int i, int &k)
```

```
{
```

```

i = 1;
k = 2;
}

```

Suppose that a main program has two integer variables x and y, which are given the value 0. Then the main program calls f(x,y); What are the values of x and y after the function f finishes?

x is still 0, but y is now 2.

Write in postfix form $5 * (9 - 7)$

5 9 7 - *

Which of the following is NOT true regarding the maze generation?

Remove a randomly chosen wall if the cells it separates are already in the same set

An array x of 100 integers is declared as

int x[100];

Local variables defined inside function body are _____ automatically at the end of function execution.

Created

Which of the following statement is NOT correct regarding Table ADT?

A table consists of several columns known as entities

In doubly linked list there is/are _____ pointer/s in each node.

Two

"new int[11]" will allocate memory for _____ integers.

11

If a tree has 20 edges/links, then the total number of nodes in the tree will be:

21

In the statement `int& a=b;`

`b` hold the address of variable `a`

Which of the following is NOT an open addressing technique to resolve collisions?

Double hashing

Consider the function X as under

```
int X (int& Value)
```

```
{
```

```
return Value;
```

```
}
```

Now `a` and `b` are integers in a calling function. Which one of the following is a valid call to the above function `X`?

`a = X(*b);`

For a perfect binary tree of height 4, What will be the sum of heights of nodes?

26

If unions are done by weight (size), the depth of any element is never greater than

$\log_2 n$

Which of the following statement is NOT true for reference variable?

References can be uninitialized.

The tree data structure is a

Non-linear data structure

In singly linked list a node consists of two parts:

Object and pointer

The simplest case in a BST to delete a node is:

When the node, that is to be deleted is a leaf node

A binary tree with 24 internal nodes has _____ external nodes.

25

$a * (b+c)-d$ is an example of _____ expression.

infix

given the values are the array representation of heap:

12 23 26 31 34 44 56 64 78 100

If we perform 4 deleteMin operations, the last element deleted is _____.

31

In _____, a programmer uses two pointers in the node, i.e. one to point to next node and the other to point to the previous node.

Linked list

Trying to remove an element from an empty stack is called _____.

Underflow of stack

For reference variables, _____ sign is used.

ampersand

Which property of equivalence relation is satisfied if we say:
Ahmad R(is related to) Ahmad

Reflexivity

Suppose there is an image segmented into pixels. Each pixel has _____ neighbour(s).

8

The most difficult case for deleting a node from Binary Search Tree is when the node to be deleted _____.

is Leaf node

When an executable program runs, it is loaded in the computer memory and becomes a _____ .

Process

If class A defines class B as its friend, then:

Class B can access private members of Class A

In the post-order traversal of a binary search tree, nodes process as:

Left-subtree, Right-subtree, Root

The expression $AB+C^*$ is called?

Postfix expression

Binary Search can be categorized into which of following?

Divide and conquer

The main use of AVL tree is:

Storing of data

Select the one FALSE statement about binary trees:

Every binary tree has at least one node.

A table consists of several columns, known as

Fields

Each operator in a postfix expression refers to the previous _____ operand(s).

One

If there are 56 internal nodes in a binary tree then how many external nodes this binary tree will have?

57

Which of the following function don't belong to the stack class?

crash()

Which of the following statement is false?

Arrays are dense lists and static data structure

During union by size method, which data structure is used to improve the balancing of tree?

Array

Non-recursive calls are faster than which of the following calls?

Recursive

Binary Search Tree violates the condition of AVL tree when any node has balance equal to

0

Each node in Binary Search Tree has

2 pointers

What will be postfix expression of the following infix expression?

*Infix Expression: $a + b * c - d$*

$abc * + d -$

The balance of a node is the result of "height of left subtree" _____ "height of right subtree".

Minus

A complete binary tree having "N" nodes consists of _____ Levels.

$\log_2 (N+1) - 1$

Consider a hash table of size seven, with starting index zero, and a hash function $(3x + 4) \bmod 7$. Assuming the hash table is initially empty, which of the following is the contents of table when the sequence 1, 3, 8, 10 is inserted into the table using closed hashing? Note that '_' denotes an empty location in the table.

1, 8, 10, _, _, _, 3

Which one of the following is known as "Last-In, First-Out" or LIFO Data Structure?

Stack

Parameters in function call are passed using,

Stack

If we singly linked list to implement list, then there is an issue that it gives difficulty when we:

Move backward in the list

Which of the following is correct about AVL Tree?

It is identical to BST except height of the left and right subtrees can differ by at most 1.

Insertion in a linked list can be done at

Front, Back and somewhere in the middle

If the height of a perfect binary tree is 4. What will be the total number of nodes in it?

31

AVL tree is linear data structure.

TRUE

During the union by size method, all the array element initialized to -1 shows:

Initial condition of tree

Consider the following pseudo code

declare a stack of characters

while (there are more characters in the word to read)

{

read a character

push the character on the stack

```

}
while ( the stack is not empty )
{
  pop a character off the stack
  write the character to the screen
}

```

What is written to the screen for the input "apples"?

selppa

Consider the linked list having data [6, 72, 35, 65, 25] stored in it. While current pointer is pointing to memory location having 72 stored in it. After calling add(4) function on the following linked list current point will point to memory location having value?

4

One difference between a queue and a stack is

Queues use two ends of the structure, stacks use only one.

next() method of List class is used to:

Moves the current position forward one element

get(?) method of list class is used to:

Get element at the given position

If we want to find median of 50 elements, then after applying buildHeap methd, how many times deleteMin method will be called?

25

In simple or singly linked list there is/are _____ pointer/s in each node.

One

If we have 1000 sets each containing a single different person.
Which of the following relation will be true on each set:

Reflexive

The total number of nodes on 10th level of a perfect binary tree are
:

1024

If there are 100 elements in a heap, and 100 deleteMin operation are performed, will get _____ list.

Sorted

In simple implementation of stack, isFull() method is used due to _____.

Limitation of array

If a tree has 50 nodes, then the total edges/links in the tree will be:

49

Following is a keyword of C++ _____ .

Delete

In level-order traversal for Binary Search Tree, at each level, we visit the nodes in _____ order.

Left-to-right

To search an element in AVL tree, it takes maximum $1.88 \log_2 n$ time.

FALSE

If we want to find 3rd minimum element from an array of elements, then after applying buildHeap method, how many times deleteMin method will be called?

3

If there are N elements in an array then the number of maximum steps needed to find an element using Binary Search is

_____.

$\log_2 N$

The _____ method of list data structure removes the element residing at the current position

Remove

Allocating and de-allocating memory for linked list nodes take _____ time than pre-allocated array.

More

In a tree, we link the nodes in such a way that it _____ a linear structure.

Forms

Longest path from root node to farthest leaf node is called _____ of tree.

Depth

In which of the following function signatures, the value of variable "num" cannot be changed in function body?

int cube(int num)

Binary search algorithm cannot be applied to _____ .

sorted linked list

The expression
if (! heap ->isFull())
check

Heap is not full

Which of the following is correct statement?

An AVL tree is identical to BST except height of the left and right subtree can differ by at most 1.

The easiest case of deleting a node from BST is the case in which the node to be deleted _____.

Is a leaf node

Every AVL is _____.

Binary Search Tree

_____ is when function is calling to itself.

Recursion

For a perfect binary tree of height h, having N nodes, the sum of heights of nodes is _____.

$N - h - 1$

The binary tree is the maximum level of its leaves (also called the depth).

Height

Which one of the following is NOT true regarding the skip list?

List Sh contains only the n special keys

Suppose we have been given the following data set for a Queue.

7 5 2 4

What will be the resultant Queue if we call enqueue(3) method?

Note that 7 is the front element whereas 4 is rear element of queue.

7 5 2 4 3

If there are N external nodes in a binary tree then what will be the no. of internal nodes in this binary tree?

N -1

int htdiff = height(root->getLeft()) _____ height(root->getRight());

The above line of code is taken from AVL insert method.

Complete it by selecting an appropriate symbol.

-

Which of the following can be used to reverse a string value?

Stack

The next field in the last node of a singly-linked list is set to

_____.

'NULL

If there are N internal nodes in a binary tree then what will be the no. of external nodes in this binary tree?

N +1

A hash function returns a _____ value.

Integer

The union operation is based on size or weight but the reducing the in-between links or path compression from _____ to the _____ is done by the find method.

nodes, root

If there are 23 external nodes in a binary tree then what will be the no. of internal nodes in this binary tree?

22

_____ in AVL is logarithmic.

Searching

In case of insertion of right inner node in BST,

we need to apply single left rotation to make it AVL tree.

Consider the following infix expression.

$7/8 + 9$

If one converts the above expression into postfix. What would be the resultant expression?

7 8 9 / +

What are the basic things associated with data structures?

All of the given

Which of the following properties are satisfied by Equivalence relationship?

Reflexive, Symmetric and Transitive

For compiler, a postfix expression is easier to evaluate than infix expression?

True

The expression $DE + H^*$ is called _____.

Hybrid expression

Consider the following expression:

$$x - y * a + b / c$$

Which of the following is a correct equivalent expression for the above?

$$x - y * a + b / c$$

In the call by _____ methodology, a copy of the object is passed to the called function.

Value

There are four cases of rotation in an _____ tree.

AVL

In singly linked list which node will keep track of starting position of the list.

Next Node

The depth of a binary tree is

Maximum level of a leaf

What will be the value of root of an AVL and BST if built from the same data?

Root of AVL is square of root of BST

In 1990, Bill Pugh proposed an enhancement on linked lists and the new data structure was termed as

Skip list

Avl tree takes maximum _____ time to search an element.

1.44 Log₂n

Can we store elements with different data types in a single array?

Yes

The _____ of a node in a binary tree is defined as the height of its left subtree minus height of its right subtree.

Balance

What is the depth of any tree if the union operation is performed by height?

O(log N)

If a hash function returns 4 for more than one value, this condition is called _____.

Collision

Which one of the following calling methods does not change the original value of the argument in the calling function?

Call by passing the value of the argument

Every AVL tree is a binary search tree.

TRUE

The difference between a "Binary Tree (BT)" and a "Binary Search Tree (BST)" is that,

In BST nodes are inserted based on the values they contain

The worst case of searching in binary search tree (BST) is:

When the data inserted in BST is sorted

Suppose there are 100 elements in an equivalence class, so initially there will be 100 trees. The collection of these trees is called

Forest

Factorial is an example of _____ function.

Recursive

Suppose a hash function returns 6 for the given value. At which index of array the value will be saved?

6

A binary tree of N nodes has _____ .

$\log_2 N$ levels

Every AVL is _____.

Binary Search Tree

If Ahmad is cousin of Ali and Ali is cousin of Asad then Ahmad is also cousin of Asad. This statement has the following property

Transitivity

What is the time complexity of binary search with iteration?

$O(\log n)$

A solution is said to be efficient if it solves the problem within its resource constraints i.e. hardware and time.

True

Searching an element in an AVL tree takes maximum _____ time (where n is number of nodes in AVL tree).

1.44 $\log_2 n$

Which of the following is not a form of expression?

Pastfix

Heap can be used to implement

Priority Queue

Sorting procedure normally takes _____ time.

$N \log N$

Consider we have performed the following operations on stack of size 5.

Push(10);

Push(20);

Push(30);

Pop();

Pop();

Push(40);

20

Which one of the following is NOT the property of equivalence relation?

Associative

What is the hash function used in linear probing?

$h_i(x) = (\text{hash}(x) + f(i)) \bmod \text{table size}$

Recursive call of a function use _____ data structure.

Stack

Left, right, info, and parent are the operation of _____ data structure.

Tree

For Binary Search Tree, we call the findMax() method as _____.

findMin(tree->getLeft())

From Operating System point of view, the recursive function calls are made with the help of _____.

Stack

Which of the following is true regarding the maze generation?

All of the given

If there are 100 elements in an equivalence class then we will have _____ sets initially.

100

given the values are the array representation of heap: 12 23 26 31 34 44 56 64 78 100

What is the 5th smallest element in the given heap?

34

Which operation of queue data structure is used to get front element from the queue and then remove it from the queue?

enqueue()

_____ is an area in computer memory that is allocated dynamically.

Heap

The tree structure is a

Non-linear data structure

Binary Tree traversal can be performed with the help of _____ .

Both Recursive and Non-recursive calls

New items are added at the _____ of the stack.

Top

Leaf node of binary search tree contains _____.

All of the given

Suppose there are a set of fruits and a set of vegetables. Both sets are _____ sets.

Disjoint

Which of the following operations returns "most recently entered value" from the stack?

Top

back() method of list class is used to:

Moves the "current" pointer to backward one element.

Stack and Queue can be implemented using _____.

Singly Link List

The array in binary search is sub divided _____.

Untill a sublist is no more divisible

Which of the following heap method lowers the value of key at position 'p' by the amount 'delta'?

decreaseKey(p,delta)

What will be the postfix expression of following infix expression?
 $A*B/C+D-E$

$A B + C D / + E -$

An efficient program executes faster and helps in _____ the usage of resources like memory and disk.

Maximizing

Array cells are _____ in computer memory.

Contiguous

Suppose there is an image of $7 * 7$, now we will have matrix of _____ rows and _____ columns.

7, 7

An array is a group of _____ memory locations.

Consecutive

for every process executing, the last part of the memory is for _____ of the program.

Heap

What will be the postfix expression of following infix expression?
 $D + E * F / G$

$DEF * G / +$

The expression
 if (! heap ->isEmpty())
 checks

Heap is not empty

_____ is used for Reference variables in C++.

&

For searching a particular number in Binary Search Tree (if it is not present), the maximum number of comparisons will be _____ comparison at each level.

2

Suppose n is the number of nodes in a complete Binary Tree, then maximum steps required for a search operation are

$\log_2 (n+1) - 1$

If we use array to implement list, then there is an issue that it gives difficulty when:

We will increase its size

Consider the following function:

```
void test_a(int n)
{
  cout << n << " ";
  if (n>0)
    test_a(n-2);
}
```

}

What is printed by the call test_a(4)?

4 2 0

Doubly Linked List always has _____ NULL pointer/s in a node.

Two

The percolateDown procedure will move the smaller value _____ and bigger value _____.

up, down

What will be the result of evaluation following expression?
 $5 + 3 * 2 / (6 - 3)$

7

_____ only removes items in reverse order as they were entered.

Stack

Linked list always contains elements that can be described as,

Self-referential

What will be postfix expression of the following infix expression?
 Infix Expression: $a + b * c - d$

$abc * + d -$

In tree, the search operation is _____ as compared to the linked list.

Very fast

Each node in doubly link list has

1 pointer

Which of the is NOT true regarding the maze generation?

Remove a randomly chosen wall if the cells it separates are already in same set

The new operation in C++ for dynamically allocating memory returns,

Pointer to the memory it has allocated

Which of the following applications may use a stack?

Parentheses balancing program

Which one of the following statement is correct?

Array size is fixed once it is created

What will be result of following postfix expression?

1 2 3 * + 2 -

4

When we compare recursive method calls and non-recursive method calls, following statement is true.

Recursion is implemented in the same way as other functioncalls are implemented

A queue where the dequeue operation does not depend upon FIFO, is called:

Priority queue

A Linear Data Structure is the data structure in which data elements are arranged in a sequence or a linear list. Which of the following is Non Linear Data Structure?

Binary Search Trees

_____ is the stack characteristic but _____ was implemented because of the size limitation of the array.

isEmpty(), isFull()

$Y = \&x[0];$

In the above statement, we get address of the first location of the array x and store it in y. Here "y" is:

lvalue

Each node in singly link list has,

1 pointer

If ahmad is boss of ehsan and ehsan is boss of umer then ahmad is also boss of umer.

The above mentioned relation is _____.

Transitive

Which of the following possible operation are performed on Table ADT?

Insert, Find, Remove

Which of the following operation returns but do not removes top value of the stack?

top

While implementing non-recursive traversal for Binary SearchTree, we need to implement _____ .

Stack

~BinarySearchTree() is a _____ .

Constructor

Which property of equivalence relation is satisfied if we say:
Ahmad is cousin of Ali and Ali is also cousin of Ahmad

Symmetry

The total number of nodes on 5th level of a perfect binary tree are :

32

In C++, we place the class interface in _____ file.

.cpp

The variables which are destroyed automatically when a function's execution ends are:

Local variables defined inside function body

If the elements are 3, 5, 1, 6 in which order they will be removed from the queue one by one

3516

The.....of a binary tree is a maximum level of its leaves (also the depth).

height

.....of the an empty AVL tree is define to be

Height

Following is the keyword of c++

Delete

In the perspective of memory organization every process expecting the first part of memory is for.....of the program.

code

The.....of a node in a binary tree is define as the height of its left subtree minus height of its right subtree.

balance

If p is pointing to a node in an existing tree then info (p)means

Return the content of node p.

Following is a keyword of c++.....

Delete

Every.....tree is a binary search tree

AVL

If a function has recursive call as last statement it is known as.....

Tail recursive

During in a order traversal call if we found a node is null it means this node will satisfy following condition

it will not have left child

In binary search tree deleting a node is easy if it is a..... node

Root

Which of the following operation returns the content of binary tree node?

Info (p)

14. 1 2 6
 5 8
 3
 4

1 2 3 4 5 6 8

in a complete binary tree the number of node of level 5 are

30

One should be careful about transient.....that are stored by reference I data structure

Objects

In a tree we link the node in such a way that it.....a linear structure

does not remain

the function call are made with the help of.....

stack

18. 15 25 45
 35 75

15 35 25 75 45

A binary is said to be a.....binary tree if every non leaf node in binary tree has non empty left and right structure

Strictly

In which of the following function signature the value of variable num cannot be changed in function body

Binary search tree

Two common method of simulations are.....are.....

Time based simulation and event based simulation

Generally there are case (s) to delete a node from BST

1

When a function calls another function the parameters and address of the calling function are put in...

Stock

Binary search tree () is a.....

Destructor

We can calculate the.....of subtree by counting its level from the bottom

Nodes

If there is a strictly complete binary tree of depth 3 the total number of nodes it will be

7

Amodel attempt to model real word phenomenon

Simulation

The process of getting the value of a variable using a pointer is called

Dereferencing

When a function calling itself is called as.....

Recursion

A.....is a tree in which every except possibly the last, is completely filled

Strict binary tree

In AVL tree during insertion, a single rotation can fix the balance in cases _____ and 4.

1

If a node is inserted in outer side of a node in binary search tree then to make it AVL tree,

We may have to apply single rotation

A zigzag rotation is performed. In Left-Left case of rotation in AVL tree.

False

The principal benefit of a linked list over a conventional array is that the order of the linked items may be_____ from the order that the data items are stored in memory.

Same

Stack.push(15) will push 15 on _____.

Top of the stack

Suppose we have the following values to be inserted in constructing AVL tree,

10,13,15,5,7,8

Tell when first rotation will take place,

After inserting the node 15

All the objects created using _____ operator have to be explicitly destroyed using the delete operator.

New

During the execution of a process, operating system constructs four things for that process. Which of the following is not part of that process?

Linked list

There are _____ cases of Rotation in AVL tree.

4

Which of the following line of code is incorrect?

Float *I = float new (not confirmed)

In simple or singly linked list there is/are _____ pointer/s in each node.

One

Suppose there are three nodes tree with three numbers 14, 4, 15. Following is not a permutation, or combination for output them.

(4, 4, 15)

The type of expression in which operator succeeds its operands is _____ expression.

Postfix

The _____ of a node in a binary tree is defined as the height of its left subtree minus height of its right subtree.

Balance

If both left and right nodes of a node are NULL then this type of node is called a _____ node.

Non leaf (not confirmed)

The function calls are made with the help of _____.

Stack

Which data structure is needed to convert infix expression to postfix expression?

Stack

In case of insertion of right inner node in BST,

Left rotation then right to make AVL

A BST generated from the data in ascending order is _____.

Linear

Which operation of queue data structure is used to get front element from the queue and then remove it from the queue?

Remove ()

In _____ various cells of memory are not located continuously.

Linked list

The _____ method of list will position the currentNode and lastCurrentNode at the start of the list.

Start

Which operation of queue data structure is used to insert an element into the Queue?

Enqueuer ()

From Operating System point of view, the recursive function calls are made with the help of _____.

Queue

We can make a lexicographic order of characters based on their _____.

ASCII values

What are the basic things associated with data structures?

All of the above

Consider the following push operations of a Stack:

Stack.push(4);

Stack.push(6);

Stack.push(5);

Stack.push(8);

If a user calls a pop() operation, then which value will be returned?

8 (stack is last in first out)

~BinarySearchTree() is a _____.

Destructor

Array cells are _____ in computer memory.

Contiguous

Leaf node of binary search tree contains _____

One Null Pointer

dequeue() operation of queue data structure is used to _____

remove an element from the front and return it

The depth of a binary tree is

Maximum level of a leaf

Which of the following operation returns but do not removes top value of the stack?

Top

Which of the following data structure is linear type?

All of above

In doubly linked list a node consists of three parts:

2 pointers and 1 object

Last node in circular linked list contains

No null pointer

A_____ model attempts to model a real-world phenomenon

Simulation

Factorial is an example of _____function.

Recursive

The balance of a node in a binary tree is defined as the height of its _____ sub tree minus height of its right sub tree.

Left

In case of insertion of left outer node in BST,

We first apply left rotation and then rotation to make it an AVL Tree.

`int htdiff = height(root->getLeft()) height(root->getRight());` The above line of code is taken from AVL insert method. Complete it by selecting an appropriate symbol.

Minus (-)

Each operator in a postfix expression refers to the previous _____ operand(s).

Two

For making Binary Search Tree for Strings we need, _____ data type.

Int

A _____ model attempts to model a real-world phenomenon

Simulation

The lifetime of a transient object cannot exceed that of the application.

True

Which one of the following calling method does not change the original value of the argument in the calling function?

Call by passing the value of the argument

In level-order traversal for Binary Search Tree, we visit the nodes at each level before proceeding to the next level, in a _____ order.

Left-to-right

In doubly linked list a node consists of three parts:

1 pointer and 2 objects

Binary Search Tree violates the condition of AVL tree when any node has balance equal to

1 or -1

To search an element in AVL tree, it takes maximum $1.88 \log_2 n$ time

True

HOW many cases of rotation are there in AVL tree?

4

_____ tree has been named after two persons Adelson-Velskii and Landis.

AVL

Which of the following is a nonlinear data structure?

Tree

AVL tree is nonlinear data structure.

True

_____ is when function is calling to itself.

Recursion

Which of the following function don't belongs to the stack class?

Crash ()

Which one is the core function call for the following function of calculating cube?

int cube(int& num)

cube (num)

Binary search algorithm cannot be applied to ____

None of the given

copy() method of list data structure _____

Set one list to be a copy of another

While implementing non-recursive traversal for Binary Search Tree, we need to implement

Stack

Following is true in case of using Recursive method calls

The code becomes very short

_____is a self-balancing tree.

Binary Search Tree

While implementing stack with an array and to achieve LIFO behavior, we used push and pop elements at

The start of the array

Generalized code written for a class is called

Structure

Army cells are _____ in computer memory

Contiguous

In singly linked list a node comprises of _____ field/s.

Two

Which of the following statement is false?

Pointers store the next data element of a list

Leaf node of binary search tree contains _____

One null Pointer

What will be the result of evaluating following expression?
 $5+3*2/(6-3)$

7

Want and de-allocating memory for linked list nodes does take
less time than pre-allocated array

MORE

Suppose you are writing a class for Node class and forgot to write
the constructor of the class, then

Compiler will automatically generate a default constructor

Factorial is an example of _____function.

Recursive

A binary tree is said to be a _____ binary tree if every non-leaf
node in a binary tree has non-empty left and right subtrees.

Strictly

Memory address is stored in

Pointer

In level-order traversal for Binary Search Tree, _____ data structure is used.

Queue

Which of the following is the correct option for priority Queue?

The type of queues that is FIFO i.e. the person who comes first should leave first.

AVL tree is linear data structure.

False

AVL tree is a binary search tree

True

The post order traversal of a binary tree is DEBFCA. Find out the pre order traversal

A

Λ

B C

Λ /

D E F

ABDCEF

In a tree, we link the nodes in such a way that it _____ a linear structure.

Does not remain

add(12) method of linked list class will:

Add 12 as value in linked list

Which of the following is the correct conversion of infix to postfix expression?

$Z+B-(D-H)/K$

ZB+DH-K/-

In singly linked list “next” field of node contains:

Address of next node

Which of the following line of code is incorrect?

The lifetime of a transient object can exceed that of the application which is accessing it.

A list is the collection of items of the _____

same type

back() method of List class is used to:

Moves the “current” pointer to backward one element.

Suppose you are writing a class for Node class and forgot to write the constructor of the class, then

_____ is the maximum height of the AVL tree.

$1.44\log_2 n$

length() method of List class is used to:

Return the length of the list

There are four cases of rotation in an _____ tree.

AVL

Allocating and de-allocating memory for linked list nodes does take _____ time than pre-allocated array.

More

_____ is utilized at the time of memory allocation in dynamic manner.

Heap

In which case of insertion we require double rotation to make the AVL tree balance.

None

The back() method decreases the value of variable current by _____.

One

In array list the worst case of removing an element is

To remove the item from start

Left, right, info, and parent are the operations of _____ data structure.

Tree

Which of the following is TRUE for search operations in a binary tree as compared to linked list or an array?

It increase the number of comparison

Which of the following is a correct statement?

An AVL tree is identical to BST except the height of left and right subtree can differ by at least 1

length() method of List class is used to:

Return the length of the list

Whenever we call a function, the compiler makes a stack, the top element of the stack is _____ of the function.

Return Address

Whenever we call a function, the compiler makes a _____ that it uses to fulfill this function call.

Stack

_____ is the major factor to see the efficiency of a program.

Time

_____ rule applies for evaluating operators of same precedence in an expression

None

The postfix form of the expression $A + B * C$ and $(A + B) * C$ will be _____.

Same

If we use array to implement list, then there is an issue that it gives difficulty when:

We will access value randomly

During in-order traversal using recursive calls, if we found a node is NULL. It means this node will satisfy following condition.

It will not have left child

Which one the following is more closer to AVL tree, (chose the best option)

Binary Search Tree

start() method of List class is used to:

Move the current pointer to first element

Consider the following infix expression. $7/8 + 9$

If one converts the above expression into postfix, what would be the resultant expression?

$78/9+$

In internal memory organization of a process, there is some area of memory for static data that holds _____ variables.

Both Static and Global

In Left-Right case of rotation in _____ tree. A double rotation is performed.

AVL

If numbers 5, 222, 4, 48 are inserted in a queue, which one will be removed first?

5

Which one is not the property of binary tree?

Sibling node should be same parent (Not Sure)

Each node in singly linked list contains _____

One Pointers

What's wrong with following loop? while((i < 10) && (i > 24)) {
}

The Condition is always false

Which one of the following is TRUE about recursion?

1. Recursive function calls consume a lot of memory.
2. **Recursion extensively uses stack memory.**
3. Recursion extensively uses stack memory.
4. Iteration is more efficient than iteration.

Which of the following is not a data structure ?

1. Linked list
2. Queue
3. stack
4. **Memory cell**

“+” is a _____ operator.

1. Unary
2. **Binary**
3. Ternary
4. None of the above

When an executable program run, it is loaded in the memory and becomes a_____.

1. Thread
2. .h file
3. **Process**
4. None of the above

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In array list the worst case of removing an element is

1. To remove an element from the end of the list
2. To remove an element from the middle of the list
3. To remove an element from the start of the list
4. We cannot remove element from an array list

The tree data structure is a

1. Linear data structure
2. Graphical data structure
3. Non-linear data structure (Not Confirm)
4. Data structure like queue

In_____, a programmer uses two pointers in the node, i.e. one to point to next node and the other to point to the previous node.

1. Linked list
2. doubly-link list
3. array
4. Structure

Each operator in a postfix expression refers to the previous _____ operand(s)

1. one
2. two
3. three
4. four

Which of the following operations returns top value of the stack?

1. push
2. pop
3. top
4. First

In the calling function, after the execution of the function called, the program continues its execution from the _____ after the function call.

1. Previous line
2. Next line
3. Beginning
4. None of the above

_____ only removes items in reverse order as they were entered.

1. Queue
2. Stack
3. Both of these
4. None of these

In the linked list implementation of the stack class, where does the push member function place the new entry on the linked list?

1. After all other entries that are greater than the new entry.
2. At the head
3. After all other entries that are smaller than the new entry.
4. At the tail

Which of the following is known as "Last-In, First-Out" or LIFO Data Structure?

1. Linked List
2. Stack
3. Queue
4. Tree

A queue is a _____ data structure, whereas a stack is a _____ data structure.

1. FIFO, LIFO

Each node in doubly link list has

1. 2
2. 3
3. 4

Whenever we call a function, the compiler makes a stack, the top element of the stack is _____ of the function

1. First argument
2. Return address
3. Last argument
4. none of the above

_____ is the stack characteristic but _____ was implemented because of the size limitation of the array.

1. isFull(),isEmpty()
2. pop(), push()
3. isEmpty() , isFull()

Stack and Queue can be implemented using _____,

1. Singly Link List
2. Binary Tree
3. Binary Search Tree
4. AVL Tree

The _____ method of list will position the currentNode and lastCurrentNode at the start of the list.

1. Remove
2. next
3. start
4. back

In array list the worst case of removing an element is

1. To remove an element from the end of the list
2. To remove an element from the middle of the list
3. To remove an element from the start of the list
4. We cannot remove element from an array list

STL is a _____ that is a part of the official standard of C++.

1. C program file
2. .h file

When an executable program run, it is loaded in the memory and becomes a_____.

1. Thread
2. .h file
3. Process
4. None of the above

Each node in doubly link list has,

1. pointer
2. pointers
3. pointers
4. pointer

A queue is a _____data structure, whereas a stack is a _____data structure.

1. FIFO, LIFO
2. LIFO,FIFO
3. both of these
4. none of these

Which one of the following is TRUE about recursion?

1. Recursive function calls consume a lot of memory.
2. Recursion extensively uses stack memory.
3. Recursion extensively uses stack memory.
4. Iteration is more efficient than iteration.

To create a _____ we link the last node with the first node in the list.

1. Double linked list
2. Circularly-linked list
3. Linked list
4. None of the above

_____ is the stack characteristic but _____ was implemented because of the size limitation of the array.

1. isFull(),isEmpty()
2. pop(), push()
3. isEmpty() , isFull()
4. push(),pop()

The principal benefit of a linked list over a conventional array is that the order of the linked items may be _____ from the order that the data items are stored in memory.

1. Same
2. Identical
3. Different
4. Equivalent

In _____ the 'next' returns false when it reaches to the last node due to the fact that the next field of the last node is set to NULL.

1. Circular linked list
2. Triple linked list
3. Singly linked list

4. None of the above

If numbers 5, 222, 4, 48 are inserted in a queue, which one will be removed first?

1. 48
2. 4
3. 333
4. 5

Which boolean expression indicates whether the numbers in two nodes (p and q) are the same. Assume that neither p nor q is null.

1. $p == q$
2. $p.data == q.data$
3. $p.nextNode == q.nextNode$
4. $p.data == q$

Local variables of a function are stored in,

1. Binary Search Tree
2. Stack
3. Queue
4. AVL Tree

A queue is a data structure where elements are

1. inserted at the front and removed from the back.
2. inserted and removed from the top.
3. inserted at the back and removed from the front.

4. inserted and removed from both ends.

A kind of expressions where the operator is present between two operands called _____ expressions.

1. Infix
2. Postfix
3. Prefix
4. None of the above

A template is a function or class that is written with a _____ data type.

1. Specific
2. Definite
3. Generic
4. None of the above.

Whenever we call a function, the compiler makes a stack, the top element of the stack is _____ of the function.

1. First argument
2. Return address
3. Last argument
4. None of the above

Which of the following can be used to reverse a string value,

1. Stack
2. Queue
3. Both of these
4. None of these

A queue is a _____ data structure, whereas a stack is a _____ data structure.

1. FIFO, LIFO
2. LIFO, FIFO
3. both of these
4. none of these

Stack and Queue can be implemented using _____,

1. Singly Link List
2. Binary Tree
3. Binary Search Tree
4. AVL Tree

Which of the following is not a data structure ?

1. Linked list
2. Stack
3. Queue
4. Memory cell

Compiler uses which one of the following in Function calls,

1. Stack

2. Queue
3. Binary Search Tree
4. AVL Tree

The principal benefit of a linked list over a conventional array is that the order of the linked items may be _____ from the order that the data items are stored in memory.

1. **Same**
2. Identical
3. Different
4. Equivalent

Doubly Linked List always has one NULL pointer.

1. True
2. **False**

We cannot remove items randomly from _____

1. Stack
2. Queue
3. Both of these
4. **None of these**

_____ is the maximum number of nodes that you can have on a stack-linked list ?

1. Zero
2. $2n$ (where n is the number of nodes in linked list)
3. **Any Number**
4. None of these

“+” is a _____ operator.

1. Unary
2. Binary
3. Ternary
4. None of the above

In _____, a programmer uses two pointers in the node, i.e. one to point to next node and the other to point to the previous node.

1. Linked list
2. doubly-link list
3. array
4. structure

A queue is a data structure where elements are

1. inserted at the front and removed from the back.
2. inserted and removed from the top.
3. inserted at the back and removed from the front.
4. inserted and removed from both ends.

_____ is the stack characteristic but _____ was implemented because of the size limitation of the array.

1. isFull(),isEmpty()
2. pop(), push()
3. isEmpty() , isFull()
4. push(),pop()

The _____ method of list will position the currentNode and lastCurrentNode at the start of the list.

1. Remove
2. Next
3. Start
4. Back

Parameters in function call are passed using,

1. Stack
2. Queue
3. Binary Search Tree
4. AVL Tree

If numbers 5, 222, 4, 48 are inserted in a queue, which one will be removed first?

1. 48
2. 4
3. 333
4. 5

For a perfect binary tree of height 4, what will be the sum of highest of node

1. 31
2. 30
3. 27
4. 26

If Ahmed is cousin of Ali and Ali is cousin of Asad then Ahmed is also cousin of Asad. This statement has the following property

1. Reflexivity

2. Symmetry
3. Transitivity
4. All of the above

Which one of the following is NOT the property of equivalence relation?

1. Reflexive
2. Symmetric
3. Transitive
4. Associative

The main reason of using heap in priority queue is

1. Improve performance
2. Code readable
3. Less code
4. Heap can't be used in priority queues

The total numbers of nodes on 10th level of perfect binary tree are

1. 256
2. 512
3. 1024
4. Can't be determined

Suppose there are 100 elements in an equivalence class, so initially there will be 100 trees, the collection of these trees is called _____.

1. Cluster
2. Class
3. Forest
4. Bunch

The percolate Down procedure will move the smaller value and bigger value.

1. Left, right
2. Right, left
3. Down, up
4. Up, down

For a perfect binary tree of height h , having N nodes, the Sum of height of nodes is

1. $N - h - 1$
2. $N - 1$
3. $N - 1 + h$
4. $N - (h - 1)$

Which of the following method is helpful in creating the heap at once?

1. Insert
2. Add
3. Update
4. percolateDown

If ahmad is boss of ehsan and ehsan is boss of umer then ahmad is also boss of umer, the above-mentioned relation is_____ .

1. Reflexive
2. Symmetry
3. Transitive
4. None of given

If we want to find median of 50 elements, then after applying builtHeap method, how many times deleteMin method will be called?

1. 5
2. 25
3. 35
4. 50

Which of the following properties are satisfied by equivalence relationship?

1. Reflexive, symmetric
2. Reflexive, transitive
3. Symmetric, transitive
4. Reflexive, symmetric and transitive

Sorting procedure normally takes____ time.

1. $N \log N$
2. $2N$
3. $N * N * N$
4. N

15. The Expression
if (! heap->isFull())
Check

1. Heap is empty
2. Heap is full
3. Heap is not empty
4. Heap is not full

The Expression
if (! heap->isEmpty())
Check

1. Heap is empty
2. Heap is full

- 3. Heap is not empty
- 4. Not a valid expression

Given the values are the array representation of heap: 12 23 26 31 34 44 56 64 78

100 If we perform 4 deleteMin operation, the last element deleted is _____ .

- 1. 31
- 2. 34
- 3. 44
- 4. 56

Which of the following heap method increase the value if key at position 'p' by the amount 'delta'?

- 1. increaseKey(p, delta)
- 2. decreaseKey(p, delta)
- 3. percolateDown(p, delta)
- 4. remove(p, delta)

Which of the following heap method lowers the value if key at position 'p' by the amount 'delta'?

- 1. increaseKey(p, delta)
- 2. decreaseKey(p, delta)
- 3. percolateDown(p, delta)
- 4. remove(p, delta)

Which property of equivalence relation is satisfied if we say:
Ahmad R(is related to)Ahmad

- 1. Reflexivity
- 2. Symmetry
- 3. Transitivity
- 4. All of Above

Given the values are the array representation of heap; 12 23 26 31 34 44 56 64 78 100 What is the 5th smallest element in the given heap?

1. 31
2. 34
3. 44
4. 56

CS301 quiz(1) file

By

<https://vuonlinehelp.blogspot.com/>

Compose by (ALI HASSAN)

If the elements are 3, 5, 1, 6 in which order they will be removed from the queue one by one

3516

The.....of a binary tree is a maximum level of its leaves (also the depth).

height

.....of the an empty AVL tree is define to be

Height

Following is the keyword of c++

Delete

In the perspective of memory organization every process expecting the first part of memory is for.....of the program.

code

The.....of a node in a binary tree is define as the height of its left

subtree minus height of its right subtree.

balance

If p is pointing to a node in an existing tree then info (p) means

Return the content of node p.

Following is a keyword of c++.....

Delete

Every.....tree is a binary search tree

AVL

If a function has recursive call as last statement it is known as.....

Tail recursive

During in a order traversal call if we found a node is null it means this node will satisfy following condition

it will not have left child

In binary search tree deleting a node is easy if it is a..... node

Root

Which of the following operation returns the content of binary tree node?

Info (p)

1 2 6

5 8

3

4

1 2 3 4 5 6 8

in a complete binary tree the number of node of level 5 are

30

One should be careful about transient.....that are stored by reference I data structure

Objects

In a tree we link the node in such a way that it.....a linear structure

does not remain

the function call are mode with the help of.....

stack

18. 15 25 45

35 75

15 35 25 75 45

A binary is said to be a.....binary tree if every non leaf node in binary tree has non empty left and right structure

Strictly

In which of the following function signature the value of variable num cannot be changed in function body

Binary search tree

Two common method of simulations are.....are.....

Time based simulation and event based simulation

Generally there are case (s)to delete a node from BST

1

When a function calls another function the parameters and address of the calling function are put in...

Stock

Binary search tree ()is a.....

Destructor

We can calculate the.....of subtree by counting its level from the bottom

Nodes

If there is a strictly complete binary tree of depth 3 the total number of nodes it will be

7

Amodel attempt to model real word phenomenon

Simulation

The process of getting the value of a variable using a pointer is called

Dereferencing

When a function calling itself is called as.....

Recursion

A.....is a tree in which every except possibly the last, is completely filled

Strict binary tree

In AVL tree during insertion, a single rotation can fix the balance in cases _____ and 4.

1

If a node is inserted in outer side of a node in binary search tree then to make it AVL tree,

We may have to apply single rotation

A zigzag rotation is performed. In Left-Left case of rotation in AVL tree.

False

The principal benefit of a linked list over a conventional array is that the order of the linked items may be_____ from the order that the data items are stored in memory.

Same

Stack.push(15) will push 15 on _____.

Top of the stack

Suppose we have the following values to be inserted in constructing AVL tree,

10,13,15,5,7,8

Tell when first rotation will take place,

After inserting the node 15

All the objects created using _____ operator have to be explicitly destroyed using the delete operator.

New

During the execution of a process, operating system constructs four things for that process. Which of the following is not part of that process?

Linked list

There are _____ cases of Rotation in AVL tree.

4

Which of the following line of code is incorrect?

Float *l = float new (not confirmed)

In simple or singly linked list there is/are _____ pointer/s in each node.

One

Suppose there are three nodes tree with three numbers 14, 4, 15. Following is not a permutation, or combination for output them.

(4, 4, 15)

The type of expression in which operator succeeds its operands is _____ expression.

Postfix

The _____ of a node in a binary tree is defined as the height of its left subtree minus height of its right subtree.

Balance

If both left and right nodes of a node are NULL then this type of node is called a _____ node.

Non leaf (not confirmed)

The function calls are made with the help of_____.

Stack

Which data structure is needed to convert infix expression to postfix expression?

Stack

In case of insertion of right inner node in BST,

Left rotation then right to make AVL

A BST generated from the data in ascending order is _____.

Linear

Which operation of queue data structure is used to get front element from the queue and then remove it from the queue?

Remove ()

In _____ various cells of memory are not located continuously.

Linked list

The _____ method of list will position the currentNode and lastCurrentNode at the start of the list.

Start

Which operation of queue data structure is used to insert an element into the Queue?

Enqueueer ()

From Operating System point of view, the recursive function calls are made with the help of _____.

Queue

We can make a lexicographic order of characters based on their _____.

ASCII values

What are the basic things associated with data structures?

All of the above

Consider the following push operations of a Stack:

Stack.push(4);

Stack.push(6);

Stack.push(5);

Stack.push(8);

If a user calls a pop() operation, then which value will be returned?

8 (stack is last in first out)

~BinarySearchTree() is a _____.

Destructor

Array cells are _____ in computer memory.

Contiguous

Leaf node of binary search tree contains _____

One Null Pointer

dequeue() operation of queue data structure is used to _____

remove an element from the front and return it

The depth of a binary tree is

Maximum level of a leaf

Which of the following operation returns but do not removes top value of the stack?

Top

Which of the following data structure is linear type?

All of above

In doubly linked list a node consists of three parts:

2 pointers and 1 object

Last node in circular linked list contains

No null pointer

A _____ model attempts to model a real-world phenomenon

Simulation

Factorial is an example of _____ function.

Recursive

The balance of a node in a binary tree is defined as the height of its _____ sub tree minus height of its right sub tree.

Left

In case of insertion of left outer node in BST,

We first apply left rotation and then rotation to make it an AVL Tree.

`int htdiff = height(root->getLeft()) - height(root->getRight());` The above line of code is taken from AVL insert method. Complete it by selecting an appropriate symbol.

Minus (-)

Each operator in a postfix expression refers to the previous _____ operand(s).

Two

For making Binary Search Tree for Strings we need, _____ data type.

Int

A _____ model attempts to model a real-world phenomenon

Simulation

The lifetime of a transient object cannot exceed that of the application.

True

Which one of the following calling method does not change the original value of the argument in the calling function?

Call by passing the value of the argument

In level-order traversal for Binary Search Tree, we visit the nodes at each level before proceeding to the next level, in a _____ order.

Left-to-right

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1 or -1

To search an element in AVL tree, it takes maximum $1.88 \log_2 n$ time

True

How many cases of rotation are there in AVL tree?

4

_____ tree has been named after two persons Adelson-Velskii and Landis.

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Which of the following is a nonlinear data structure?

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AVL tree is nonlinear data structure.

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_____ is when function is calling to itself.

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Which of the following function don't belongs to the stack class?

Crash ()

Which one is the cored function call for the following function of calculating cube?

```
int cube(int& num)
```

cube (num)

Binary search algorithm cannot be applied to ____

None of the given

copy() method of list data structure _____

Set one list to be a copy of another

While implementing non-recursive traversal for Binary Search Tree, we need to implement

Stack

Following is true in case of using Recursive method calls

The code becomes very short

_____is a self-balancing tree.

Binary Search Tree

While implementing stack with an array and to achieve LIFO behavior, we used push and pop elements at

The start of the array

Generalized code written for a class is called

Structure

Array cells are _____ in computer memory

Contiguous

In singly linked list a node comprises of _____ field/s.

Two

Which of the following statement is false?

Pointers store the next data element of a list

Leaf node of binary search tree contains _____

One null Pointer

What will be the result of evaluating following expression?

$5 + 3 * 2 / (6 - 3)$

7

Want and de-allocating memory for linked list nodes does take less time than pre-allocated array

MORE

Suppose you are writing a class for Node class and forgot to write the constructor of the class, then

Compiler will automatically generate a default constructor

Factorial is an example of _____ function.

Recursive

A binary tree is said to be a _____ binary tree if every non-leaf node in a binary tree has non-empty left and right subtrees.

Strictly

Memory address is stored in

Pointer

In level-order traversal for Binary Search Tree, _____ data structure is used.

Queue

Which of the following is the correct option for priority Queue?

The type of queues that is FIFO i.e. the person who comes first should leave first.

AVL tree is linear data structure.

False

AVL tree is a binary search tree

True

The post order traversal of a binary tree is DEBFCA. Find out the pre order traversal

A

Λ

B C

$\wedge /$

D E F

ABDCEF

In a tree, we link the nodes in such a way that it _____ a linear structure.

Does not remain

add(12) method of linked list class will:

Add 12 as value in linked list

Which of the following is the correct conversion of infix to postfix expression?

$Z+B-(D-H)/K$

ZB+DH-K/-

In singly linked list "next" field of node contains:

Address of next node

Which of the following line of code is incorrect?

The lifetime of a transient object can exceed that of the application which is accessing it.

A list is the collection of items of the _____

same type

back() method of List class is used to:

Moves the "current" pointer to backward one element.

Suppose you are writing a class for Node class and forgot to write the constructor of the class, then

_____ is the maximum height of the AVL tree.

$1.44\log_2 n$

length() method of List class is used to:

Return the length of the list

There are four cases of rotation in an _____ tree.

AVL

Allocating and de-allocating memory for linked list nodes does take _____ time than pre-allocated array.

More

_____ is utilized at the time of memory allocation in dynamic manner.

Heap

In which case of insertion we require double rotation to make the AVL tree balance.

None

The back() method decreases the value of variable current by _____.

One

In array list the worst case of removing an element is

To remove the item from start

Left, right, info, and parent are the operations of _____ data structure.

Tree

Which of the following is TRUE for search operations in a binary tree as compared to linked list or an array?

It increase the number of comparison

Which of the following is a correct statement?

An AVL tree is identical to BST except the height of left and right subtree can differ by at least 1

length() method of List class is used to:

Return the length of the list

Whenever we call a function, the compiler makes a stack, the top element of the stack is _____ of the function.

Return Address

Whenever we call a function, the compiler makes a _____ that it uses to fulfill this function call.

Stack

_____ is the major factor to see the efficiency of a program.

Time

_____ rule applies for evaluating operators of same precedence in an expression

None

The postfix form of the expression $A + B * C$ and $(A + B) * C$ will be _____.

Same

If we use array to implement list, then there is an issue that it gives difficulty when:

We will access value randomly

During in-order traversal using recursive calls, if we found a node is NULL. It means this node will satisfy following condition.

It will not have left child

Which one the following is more closer to AVL tree, (chose the best option)

Binary Search Tree

start() method of List class is used to:

Move the current pointer to first element

Consider the following infix expression. $7/8 + 9$

If one converts the above expression into postfix, what would be the resultant expression?

78/9+

In internal memory organization of a process, there is some area of memory for static data that holds _____ variables.

Both Static and Global

In Left-Right case of rotation in _____ tree. A double rotation is performed.

AVL

If numbers 5, 222, 4, 48 are inserted in a queue, which one will be removed first?

5

Which one is not the property of binary tree?

Sibling node should be same parent (Not Sure)

Each node in singly linked list contains_____

One Pointers

What's wrong with following loop? `while( (i < 10) && (i > 24)) { }`

The Condition is always false

Which one of the following is TRUE about recursion?

1. Recursive function calls consume a lot of memory.
2. Recursion extensively uses stack memory.
3. Recursion extensively uses stack memory.
4. Iteration is more efficient than iteration.

Which of the following is not a data structure ?

1. Linked list

2. Queue
3. stack
4. Memory cell

“+” is a _____operator.

1. Unary
2. Binary
3. Ternary
4. None of the above

When an executable program run, it is loaded in the memory and becomes a_____.

1. Thread
2. .h file
3. Process
4. None of the above

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In array list the worst case of removing an element is

1. To remove an element from the end of the list
2. To remove an element from the middle of the list
3. To remove an element from the start of the list
4. We cannot remove element from an array list

The tree data structure is a

1. Linear data structure

2. Graphical data structure
3. Non-linear data structure (Not Confirm)
4. Data structure like queue

In_____, a programmer uses two pointers in the node, i.e. one to point to next node and the other to point to the previous node.

1. Linked list
2. doubly-link list
3. array
4. Structure

Each operator in a postfix expression refers to the previous _____ operand(s)

1. one
2. two
3. three
4. four

Which of the following operations returns top value of the stack?

1. push
2. pop
3. top
4. First

In the calling function, after the execution of the function called, the program continues its execution from the _____ after the function call.

1. Previous line
2. Next line
3. Beginning
4. None of the above

_____ only removes items in reverse order as they were entered.

1. Queue
2. Stack
3. Both of these
4. None of these

In the linked list implementation of the stack class, where does the push member function place the new entry on the linked list?

1. After all other entries that are greater than the new entry.
2. At the head
3. After all other entries that are smaller than the new entry.
4. At the tail

Which of the following is known as "Last-In, First-Out" or LIFO Data Structure?

1. Linked List
2. Stack
3. Queue
4. Tree

A queue is a _____data structure, whereas a stack is a _____data structure.

1. FIFO, LIFO

Each node in doubly link list has

1. 2
2. 3
3. 4

Whenever we call a function, the compiler makes a stack, the top element of the stack is _____ of the function

1. First argument
2. Return address
3. Last argument
4. none of the above

_____ is the stack characteristic but _____ was implemented because of the size limitation of the array.

1. isFull(),isEmpty()

2. pop(), push()
3. isEmpty() , isFull()

Stack and Queue can be implemented using _____,

1. Singly Link List
2. Binary Tree
3. Binary Search Tree
4. AVL Tree

The _____ method of list will position the currentNode and lastCurrentNode at the start of the list.

1. Remove
2. next
3. start
4. back

In array list the worst case of removing an element is

1. To remove an element from the end of the list

2. To remove an element from the middle of the list
3. To remove an element from the start of the list
4. We cannot remove element from an array list

STL is a _____ that is a part of the official standard of C++.

1. C program file

2. .h file

When an executable program run, it is loaded in the memory and becomes a_____.

1. Thread

2. .h file

3. Process

4. None of the above

Each node in doubly link list has,

1. pointer

2. pointers

3. pointers

4. pointer

A queue is a _____data structure, whereas a stack is a _____data structure.

1. FIFO, LIFO

2. LIFO,FIFO

3. both of these

4. none of these

Which one of the following is TRUE about recursion?

1. Recursive function calls consume a lot of memory.

2. Recursion extensively uses stack memory.

3. Recursion extensively uses stack memory.
4. Iteration is more efficient than iteration.

To create a _____ we link the last node with the first node in the list.

1. Double linked list
2. Circularly-linked list
3. Linked list
4. None of the above

_____ is the stack characteristic but _____ was implemented because of the size limitation of the array.

1. isFull(),isEmpty()
2. pop(), push()

3. isEmpty() , isFull()

4. push(),pop()

The principal benefit of a linked list over a conventional array is that the order of the linked items may be_____ from the order that the data items are stored in memory.

1. Same

2. Identical

3. Different

4. Equivalent

In_____ the 'next' returns false when it reaches to the last node due to the fact that the next field of the last node is set to NULL.

1. Circular linked list

2. Triple linked list

3. Singly linked list

4. None of the above

If numbers 5, 222, 4, 48 are inserted in a queue, which one will be removed first?

1. 48

2. 4

3. 333

4. 5

Which boolean expression indicates whether the numbers in two nodes (p and q) are the same. Assume that neither p nor q is null.

1. `p == q`

2. `p.data == q.data`

3. `p.nextNode == q.nextNode`

4. `p.data == q`

Local variables of a function are stored in,

1. Binary Search Tree

2. Stack

3. Queue

4. AVL Tree

A queue is a data structure where elements are

1. inserted at the front and removed from the back.

2. inserted and removed from the top.

3. inserted at the back and removed from the front.

4. inserted and removed from both ends.

A kind of expressions where the operator is present between two

operands called _____expressions.

1. Infix

2. Postfix

3. Prefix

4. None of the above

A template is a function or class that is written with a _____data type.

1. Specific

2. Definite

3. Generic

4. None of the above.

Whenever we call a function, the compiler makes a stack, the top element of the stack is _____ of the function.

1. First argument

2. Return address

3. Last argument

4. None of the above

Which of the following can be used to reverse a string value,

1. Stack

2. Queue

3. Both of these

4. None of these

A queue is a _____data structure, whereas a stack is a _____data structure.

1. FIFO, LIFO

2. LIFO,FIFO
3. both of these
4. none of these

Stack and Queue can be implemented using _____,

1. Singly Link List
2. Binary Tree
3. Binary Search Tree
4. AVL Tree

Which of the following is not a data structure ?

1. Linked list
2. Stack
3. Queue
4. Memory cell

Compiler uses which one of the following in Function calls,

1. Stack
2. Queue
3. Binary Search Tree
4. AVL Tree

The principal benefit of a linked list over a conventional array is that the order of the linked items may be_____ from the order that the data items are stored in memory.

1. Same
2. Identical
3. Different
4. Equivalent

Doubly Linked List always has one NULL pointer.

1. True
2. False

We cannot remove items randomly from _____

1. Stack
2. Queue
3. Both of these
4. None of these

_____ is the maximum number of nodes that you can have on a stack-linked list ?

1. Zero
2. $2n$ (where n is the number of nodes in linked list)
3. Any Number
4. None of these

“+” is a _____operator.

1. Unary
2. Binary
3. Ternary
4. None of the above

In_____, a programmer uses two pointers in the node, i.e. one to point to next node and the other to point to the previous node.

1. Linked list
2. doubly-link list
3. array
4. structure

A queue is a data structure where elements are

1. inserted at the front and removed from the back.
2. inserted and removed from the top.

3. inserted at the back and removed from the front.

4. inserted and removed from both ends.

_____ is the stack characteristic but _____ was implemented because of the size limitation of the array.

1. isFull(),isEmpty()

2. pop(), push()

3. isEmpty() , isFull()

4. push(),pop()

The _____ method of list will position the currentNode and lastCurrentNode at the start of the list.

1. Remove

2. Next

3. Start

4. Back

Parameters in function call are passed using,

-
1. Stack
 2. Queue
 3. Binary Search Tree
 4. AVL Tree

If numbers 5, 222, 4, 48 are inserted in a queue, which one will be removed first?

1. 48
2. 4
3. 333
4. 5

For a perfect binary tree of height 4, what will be the sum of highest of node

1. 31
2. 30
3. 27

4. 26

If Ahmed is cousin of Ali and Ali is cousin of Asad then Ahmed is also cousin of Asad. This statement has the following property

1. Reflexivity
2. Symmetry
3. Transitivity
4. All of the above

Which one of the following is NOT the property of equivalence relation?

1. Reflexive
2. Symmetric
3. Transitive
4. Associative

The main reason of using heap in priority queue is

1. Improve performance

2. Code readable
3. Less code
4. Heap can't be used in priority queues

The total numbers of nodes on 10th level of perfect binary tree are

1. 256
2. 512
3. 1024
4. Can't be determined

Suppose there are 100 elements in an equivalence class, so initially there will be 100 trees, the collection of these trees is called _____.

1. Cluster
2. Class
3. Forest
4. Bunch

The percolate Down procedure will move the smaller value and

bigger value.

1. Left, right
2. Right, left
3. Down, up
4. Up, down

For a perfect binary tree of height h , having N nodes, the Sum of height of nodes is

1. $N - h - 1$
2. $N - 1$
3. $N - 1 + h$
4. $N - (h - 1)$

Which of the following method is helpful in creating the heap at once?

1. Insert
2. Add
3. Update

4. percolateDown

If ahmad is boss of ehsan and ehsan is boss of umer then ahmad is also boss of umer, the above-mentioned relation is_____ .

1. Reflexive
2. Symmetry
3. Transitive
4. None of given

If we want to find median of 50 elements, then after applying builtHeap method, how many times deleteMin method will be called?

1. 5
2. 25
3. 35
4. 50

Which of the following properties are satisfied by equivalence relationship?

1. Reflexive, symmetric

2. Reflexive, transitive

3. Symmetric, transitive

4. Reflexive, symmetric and transitive

Sorting procedure normally takes___ time.

1. $N \log N$

2. $2N$

3. $N*N*N$

4. N

15. The Expression

if (! heap->isFull())

Check

1. Heap is empty

2. Heap is full

3. Heap is not empty

4. Heap is not full

The Expression

if (! heap->isEmpty())

Check

1. Heap is empty
2. Heap is full
3. Heap is not empty
4. Not a valid expression

Given the values are the array representation of heap: 12 23 26 31
34 44 56 64 78

100 If we perform 4 deleteMin operation, the last element deleted is_____ .

1. 31

2. 34

3. 44

4. 56

Which of the following heap method increase the value if key at

position 'p' by the amount 'delta'?

1. increaseKey(p, delta)
2. decreaseKey(p, delta)
3. percolateDown(p, delta)
4. remove(p, delta)

Which of the following heap method lowers the value if key at position 'p' by the amount 'delta'?

1. increaseKey(p, delta)
2. decreaseKey(p, delta)
3. percolateDown(p, delta)
4. remove(p, delta)

Which property of equivalence relation is satisfied if we say:
Ahmad R(is related to)Ahmad

1. Reflexivity
2. Symmetry

3. Transitivity

4. All of Above

Given the values are the array representation of heap; 12 23 26 31 34 44 56 64 78 100 What is the 5th smallest element in the given heap?

1. 31

2. 34

3. 44

4. 56

CS301 Quiz 1 Solution Fall 2020

Quiz File 1

Question No 1

If there are 23 external nodes in a binary tree then what will be the no. of internal nodes in this binary tree?

23

Question No 2

In which of the following tree, parent node has key greater than or equal to its both children?

Max heap

Question No 3

See the below code and fill the appropriate answer for ? sign.

```
void fastInorder(TreeNode* p)
{
    while((p=nexInorder(p)) != ? )
        cout << p->getInfo();
}
```

dummy

Question No 4

In a threaded binary tree which nodes have NULL child pointers,

All leaf nodes

Question No 5

Which of the following statement concerning heaps is NOT true?

Traversing a heap in order provides access to the data in numeric or alphabetical order.

Question No 6

Consider a min heap, represented by the following array:

3,4,6,7,5,10

After inserting a node with value 1. Which of the following is the updated min heap?

3,4,6,7,5,10,1

Question No 7

Which of the following statement is correct?

A Threaded Binary Tree is a binary tree in which every node that does not have a right child has a THREAD (in actual sense, a link) to its PREORDER successor.

Question No 8

A complete binary tree is a tree that is _____ filled, with the possible exception of the bottom level.

Completely

Question No 9

Consider a binary tree, represented by the following array:

10,7,9,5,2,1,6,3,4

This is a _____.

Max heap

Question No 10

When a complete binary tree, represented by an array then for any array element at position i , the right child is at position _____.

$2i+1$

Quiz File 2

Question No 1

We implement the heap by _____.

Complete binary tree

Question No 2

Consider a max heap, represented by the following array:

40,30,20,10,15,16,17,8,4

After inserting a node with value 35. Which of the following is the updated max heap?

40,30,20,10,15,16,17,8,4,35

Question No 3

A complete binary tree of height 3 has between _____ nodes.

8 to 15

Question No 4

It is necessary for Huffman encoding tree to be,

Binary Tree

Question No 5

If there are N internal nodes in a binary tree then what will be the no. of external nodes in this binary tree?

N+1

Question No 6

Which of the following statement concerning heaps is NOT true?

A heap can be used to sort data.

Question No 7

Which of the following statement is correct?

A Threaded Binary Tree is a binary tree in which every node that does not have a right child has a THREAD (in actual sense, a link) to its PREORDER successor.

Question No 8

When a complete binary tree, represented by an array then for any array element at position i, the right child is at position _____ .

2i+1

Question No 9

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3,4,6,7,5,10

After inserting a node with value 1. Which of the following is the updated min heap?

3,4,6,7,5,10,1

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Question No 10

Consider a binary tree, represented by the following array:

10,7,9,5,2,1,6,3,4

This is a _____.

Max heap

VUStudents.pk

For More Visit: VUStudents.pk

A queue is a _____ data structure, whereas a stack is a _____ data structure.

FIFO, LIFO

Each node in doubly link list has

2

3

4

Whenever we call a function, the compiler makes a stack, the top element of the stack is _____ of the function

First argument

Return address

Last argument

none of the above

To create a _____ we link the last node with the first node in the list.

Double linked list

Circularly-linked list

_____ is the stack characteristic but _____ was implemented because of the size limitation of the array.

isFull(), isEmpty()

pop(), push()

isEmpty() , isFull()

Stack and Queue can be implemented using _____,

Singly Link List

Binary Tree

Binary Search Tree

AVL Tree

The _____ method of list will position the currentNode and lastCurrentNode at the start of the list.

Remove

next

start

back

In array list the worst case of removing an element is

To remove an element from the end of the list

To remove an element from the middle of the list

To remove an element from the start of the list

We cannot remove element from an array list

STL is a _____ that is a part of the official standard of C++.

C program file

.h file

When an executable program runs, it is loaded in the memory and becomes a_____.

Select correct option:

Thread

.h file

Process

None of the above

Each node in doubly linked list has,

Select correct option:

1 pointer

2 pointers

3 pointers

4 pointers

Which of the following operations returns top value of the stack?

Select correct option:

push

pop

top

first

A queue is a _____ data structure, whereas a stack is a _____ data structure.

Select correct option:

FIFO, LIFO

LIFO, FIFO

both of these

none of these

Which one of the following is TRUE about recursion?

Select correct option:

Recursive function calls consume a lot of memory.

Recursion extensively uses stack memory.

Recursion extensively uses stack memory.

Iteration is more efficient than iteration.

To create a _____ we link the last node with the first node in the list.

Select correct option:

Double linked list

Circularly-linked list

Linked list

None of the above

_____ is the stack characteristic but _____ was implemented because of the size limitation of the array.

Select correct option:

isFull(),isEmpty()

pop(), push()

isEmpty() , isFull()

push(),pop()

The principal benefit of a linked list over a conventional array is that the order of the linked items may be_____ from the order that the data items are stored in memory.

Select correct option:

Same

Identical

Different

Equivalent

In_____ the 'next' returns false when it reaches to the last node due to the fact that the next field of the last node is set to NULL.

Select correct option:

Circular linked list

Triple linked list

Singly linked list

None of the above

If numbers 5, 222, 4, 48 are inserted in a queue, which one will be removed first?

Select correct option:

48

4

333

5

Which boolean expression indicates whether the numbers in two nodes (p and q) are the same. Assume that neither p nor q is null.

Select correct option:

$p == q$

$p.data == q.data$

$p.nextNode == q.nextNode$

$p.data == q$

Local variables of a function are stored in,

Select correct option:

Binary Search Tree

Stack

Queue

AVL Tree

A queue is a data structure where elements are

Select correct option:

inserted at the front and removed from the back.

inserted and removed from the top.

inserted at the back and removed from the front.

inserted and removed from both ends.

A kind of expressions where the operator is present between two operands called _____ expressions.

Select correct option:

Infix

Postfix

Prefix

None of the above

A template is a function or class that is written with a _____ data type.

Select correct option:

Specific

Definite

Generic

None of the above.

Whenever we call a function, the compiler makes a stack, the top element of the stack is _____ of the function.

Select correct option:

First argument

Return address

Last argument

None of the above

Which of the following can be used to reverse a string value,

Select correct option:

Stack
Queue
Both of these
None of these

A queue is a _____ data structure, whereas a stack is a _____ data structure.

Select correct option:

FIFO, LIFO
LIFO, FIFO
both of these
none of these

Stack and Queue can be implemented using _____,

Select correct option:

Singly Link List
Binary Tree
Binary Search Tree
AVL Tree

"+" is a _____ operator.

Select correct option:

Unary
Binary
Ternary
None of the above

Which of the following is not a data structure ?

Select correct option:

Linked list

Stack

Queue

Memory cell

Compiler uses which one of the following in Function calls,
Select correct option:

Stack

Queue

Binary Search Tree

AVL Tree

The principal benefit of a linked list over a conventional array is that the order of the linked items may be_____ from the order that the data items are stored in memory.

Select correct option:

Same

Identical

Different

Equivalent

Doubly Linked List always has one NULL pointer.

Select correct option:

True

False

We can not remove items randomly from _____

Select correct option:

Stack

Queue

Both of these

None of these

_____ is the maximum number of nodes that you can have on a stack-linked list ?

Select correct option:

Zero

$2n$ (where n is the number of nodes in linked list)

Any Number

None of these

"+" is a _____ operator.

Select correct option:

Unary

Binary

Ternary

None of the above

: In _____, a programmer uses two pointers in the node, i.e. one to point to next node and the other to point to the previous node.

Select correct option:

Linked list

doubly-link list

array
structure

A queue is a data structure where elements are
Select correct option:

inserted at the front and removed from the back.
inserted and removed from the top.
inserted at the back and removed from the front.
inserted and removed from both ends.

: _____ is the stack characteristic but _____ was implemented because of
the size limitation of the array.

Select correct option:

isFull(),isEmpty()
pop(), push()
isEmpty() , isFull()
push(),pop()

The _____ method of list will position the currentNode and lastCurrentNode
at the start of the list.

Select correct option:

Remove
Next
Start
Back

: Parameters in function call are passed using,
Select correct option:

Stack

Queue

Binary Search Tree

AVL Tree

If numbers 5, 222, 4, 48 are inserted in a queue, which one will be removed first?

Select correct option:

48

4

333

5

CS301 Quiz No 1 Fall 2015

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CS301 QUIZ 1 SOLVED

Provide by VU Answer

1. For a perfect binary tree of height 4, what will be the sum of highest of node

a) 31

b) 30

c) 27

d) 26

2. If Ahmed is cousin of Ali and Ali is cousin of Asad then Ahmed is also cousin of Asad. This statement has the following property

a) Reflexivity

b) Symmetry

c) Transitivity

d) All of the above

3. Which one of the following is NOT the property of equivalence relation?

a) Reflexive

b) Symmetric

c) Transitive

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d) Associative

4. The main reason of using heap in priority queue is

a) Improve performance

b) Code readable

c) Less code

d) Heap can't be used in priority queues

5. The total number of nodes on 10th level of perfect binary tree are

a) 256

b) 512

c) 1024

d) Can't be determined

6. Suppose there are 100 elements in an equivalence class, so initially there will be 100 trees, the collection of these trees is called_____.

a) Cluster

b) Class

c) Forest

d) Bunch

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7. The percolate Down procedure will move the smaller value_____and bigger value.

a) Left, right

b) Right, left

c) Down, up

d) Up, down

8. For a perfect binary tree of height h , having N nodes, the Sum of height of nodes is __

a) $N - h - 1$

b) $N - 1$

c) $N - 1 + h$

d) $N - (h - 1)$

10. Which of the following method is helpful in creating the heap at once?

a) Insert

b) Add

c) Update

d) percolateDown

11. If ahmad is boss of ehsan and ehsan is boss of umer then ahmad is also boss of umer, the above-mentioned relation is_____.

a) Reflexive

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b) Symmetry

c) Transitive

d) None of given

12. If we want to find median of 50 elements, then after applying builtHeap method, how manytime deleteMin method will be called?

a) 5

b) 25

c) 35

d) 50

13. Which of the following properties are satisfied by equivalence relationship?

a) Reflexive, symmetric

b) Reflexive, transitive

c) Symmetric, transitive

d) Reflexive, symmetric and transitive

14. Sorting procedure normally takes____time.

a) $N \log N$

b) $2N$

c) $N*N*N$

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d) N

15. The Expression

if (! heap->isFull())

Check

a) Heap is empty

b) Heap is full

c) Heap is not empty

d) Heap is not full

16. The Expression

if (! heap->isEmpty())

Check

a) Heap is empty

b) Heap is full

c) Heap is not empty

d) Not a valid expression

17. Given the values are the array representation of heap: 12 23 26 31 34 44 56 64 78 100

If we perform 4 deleteMin operation, the last element deleted is ____.

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a) 31

b) 34

c) 44

d) 56

18. Which of the following heap method increase the value if key at position 'p' by the amount 'delta'?

a) increaseKey(p, delta)

b) decreaseKey(p, delta)

c) percolateDown(p, delta)

d) remove(p, delta)

19. Which of the following heap method lowers the value if key at position 'p' by the amount 'delta'?

a) increaseKey(p, delta)

b) decreaseKey(p, delta)

c) percolateDown(p, delta)

d) remove(p, delta)

20. Which property of equivalence relation is satisfied if we say:

Ahmad R(is related to)Ahmad

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a) Reflexivity

- b) Symmetry
- c) Transitivity
- d) All of Above

21. Given the values are the array representation of heap; 12 23 26 31 34 44 56 64 78 100

What is the 5th smallest element in the given heap?

a) 31

b) 34

c) 44

d) 56

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CS301- Data Structures

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MC100401285

Moaaz.pk@gmail.com *MC100401285@gmail.com*

PSMD01

mc110200885

virtualcheater@gmail.com

VDGK01

mc110200030

awais4you@gmail.com

PLHR25

CS301-MIDTERM SOLVED MCQS WITH REFERENCES



Muhammad Moaaz Siddiq MCS (2nd)
mc100401285@gmail.com
Campus:- Institute of E-Learning &
Modern Studies (IEMS) Samundar
(PSMD01)

Muhammad Irfan Farid
MCS(2nd)
virtualcheater@gmail.com
Virtual University Campus . D.G. Khan
(VDGK01)

Muhammad Awais- MCS(2nd)
awais4you@gmail.com
Campus : -Abbas Institute Of
Modern Studies(AIMS)Lahore
(PLHR25)

Muhammad Moaaz Siddiq MCS (2nd)
mc100401285@gmail.com
Campus:- Institute of E-Learning &
Modern Studies (IEMS) Samundar
(PSMD01)

Muhammad Irfan Farid
MCS(2nd)
virtualcheater@gmail.com
Virtual University Campus . D.G. Khan
(VDGK01)

Muhammad Awais- MCS(2nd)
awais4you@gmail.com
Campus : -Abbas Institute Of
Modern Studies(AIMS)Lahore
(PLHR25)

MIDTERM EXAMINATION

Spring 2010

CS301- Data Structures

Question No: 1 (Marks: 1) - Please choose one

A queue where the de-queue operation depends not on FIFO, is called a priority queue

- ▶ False
- ▶ **True (Page 101)**

Question No: 2 (Marks: 1) - Please choose one

The data of the problem is of 2GB and the hard disk is of 1GB capacity, to solve this problem we should

- ▶ Use better data structures
- ▶ **Increase the hard disk space (Page 5)**
- ▶ Use the better algorithm
- ▶ Use as much data as we can store on the hard disk

Question No: 3 (Marks: 1) - Please choose one

Consider the function X as under

```
int X (int& Value)
{
    return Value;
}
```

Now a and b are integers in a calling function. Which one of the following is a valid call to the above function X.

- ▶ a = X (b) ;
- ▶ **a = X (&b) ;**
- ▶ a = X (*b) ;
- ▶ None of the given options

Here function argument passing by reference method is used, so when we call a function we will give the variable reference as parameter.

Question No: 4 (Marks: 1) - Please choose one

In the call by value methodology, a copy of the object is passed to the called function.

- ▶ False
- ▶ **True (Page 202)**

Question No: 5 (Marks: 1) - Please choose one

The tree data structure is a

- ▶ Linear data structure
- ▶ **Non-linear data structure (Page 112)**
- ▶ Graphical data structure
- ▶ Data structure like queue

Question No: 6 (Marks: 1) - Please choose one

When should you use a **const reference** parameter?

- ▶ Whenever the parameter has huge size.
- ▶ **Whenever the parameter has huge size, the function changes the parameter within its body, and you do NOT want these changes to alter the actual argument.**
- ▶ Whenever the parameter has huge size, the function changes the parameter within its body, and you DO want these changes to alter the actual argument.
- ▶ Whenever the parameter has huge size, and the function does not change the parameter within its body.

Declaring a parameter as a const simply means that the function can't change the value of its parameters.

Question No: 7 (Marks: 1) - Please choose one

Here is the start of a C++ class declaration:

```
class foo
{
public:
    void x(foo f);
    void y(const foo f);
    void z(foo f) const;
...
}
```

Which of the three member functions can alter the PRIVATE member variables of the foo object that activates the function?

- ▶ Only x can alter the private member variables of the object that activates the function.
- ▶ Only y can alter the private member variables of the object that activates the function.
- ▶ Only z can alter the private member variables of the object that activates the function.
- ▶ **Two of the functions can alter the private member variables of the object that activates the function.**

Only the x and y can alter the private member variable of the foo class object. Last Option is more correct but not exact. In the last option the two function name are not mentioned

Question No: 8 (Marks: 1) - Please choose one

What is the maximum depth of recursive calls a function may make?

- ▶ 1
- ▶ 2
- ▶ n (where n is the argument)
- ▶ **There is no fixed maximum**

Question No: 9 (Marks: 1) - Please choose one

Suppose n is the number of nodes in a complete Binary Tree then maximum steps required for a search operation are,

- ▶ **$\log_2 (n+1) - 1$ (Page 139)**
- ▶ $\log_2 (n+1)$
- ▶ $\log_2 (n) - 1$
- ▶ $\log_2 (n)$

Question No: 10 (Marks: 1) - Please choose one

In the linked list implementation of the stack class, where does the push member function places the new entry on the linked list?

- ▶ **At the head (Page 53)**
- ▶ At the tail
- ▶ After all other entries that are greater than the new entry.
- ▶ After all other entries that are smaller than the new entry.

Question No: 11 (Marks: 1) - Please choose one

Suppose we have a *circular* array implementation of the queue class, with ten items in the queue stored at data[2] through data[11]. The CAPACITY is 42, i.e., the array has been declared to be of size 42. Where does the push member function place the new entry in the array?

- ▶ data[1]
- ▶ data[2]
- ▶ data[11]
- ▶ **data[12]**

Question No: 12 (Marks: 1) - Please choose one

The expression $AB+C^*$ is called?

- ▶ Prefix expression
- ▶ **Postfix expression (Page 70)**
- ▶ Infix expression
- ▶ None of these

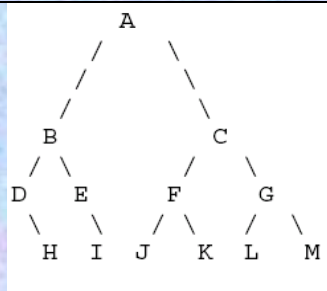
Question No: 13 (Marks: 1) - Please choose one

_____ is a binary tree where every node has a value, every node's left subtree contains only values less than or equal to the node's value, and every node's right subtree contains only values that are greater then or equal?

- ▶ Strictly Binary Tree
- ▶ **Binary Search tree** [Click here for detail](#)
- ▶ AVL tree
- ▶ All of these

Question No: 14 (Marks: 1) - Please choose one

Consider the following binary search tree (BST):



If node A in the BST is deleted, which two nodes are the candidates to take its place?

- ▶ J and I
- ▶ H and E
- ▶ D and E
- ▶ **L and M**

Question No: 15 (Marks: 1) - Please choose one

Let's call the node as that requires re-balancing. Consider the two cases given below:

- 1) An insertion into left sub tree of the left child of a
- 2) An insertion into right sub tree of the right child of a.

Which of the following statement is correct about these two cases?

- 1) The insertion occurs outside (i.e., left-left or right-right) in cases 1 and 2 single rotation can fix the balance in these two cases.
- 2) The insertion occurs inside ((i.e., left-left or right-right) in cases 1 and 2. single rotation cannot fix the balance in these two cases

Question No: 16 (Marks: 1) - Please choose one

We access elements in AVL Tree in,

- ▶ Linear way only
- ▶ **Non Linear way only**
- ▶ Both linear and non linear ways
- ▶ None of the given options.

Question No: 17 (Marks: 2)

AVL Tree is,

- ▶ **Non Linear data structure** [Click here for detail](#)
- ▶ Linear data structure
- ▶ Hybrid data structure (Mixture of Linear and Non Linear)
- ▶ None of the given options.

MIDTERM EXAMINATION

Spring 2010

Question No: 1 (Marks: 1) - Please choose one

Which one of the following statement is NOT correct .

- ▶ **In linked list the elements are necessarily to be contiguous** [Click here for detail](#)
- ▶ In linked list the elements may locate at far positions in the memory
- ▶ In linked list each element also has the next to it
- ▶ In an array the elements are contiguous

Question No: 2 (Marks: 1) - Please choose one

Each operator in a postfix expression refers to the previous _____ operand(s).

- ▶ One
- ▶ **Two (Page 67)**
- ▶ Three
- ▶ Four

Question No: 3 (Marks: 1) - Please choose one

Which one of the following calling methods does not change the original value of the argument in the calling function?

- ▶ None of the given options
- ▶ **Call by passing the value of the argument** [Click here for detail](#)
- ▶ Call by passing reference of the argument
- ▶ Call by passing the address of the argument

Question No: 4 (Marks: 1) - Please choose one

A tree is an AVL tree if

- ▶ Any one node fulfills the AVL condition
- ▶ At least half of the nodes fulfill the AVL condition
- ▶ **All the nodes fulfill the AVL condition (Page 213)**
- ▶ None of the given options

Question No: 5 (Marks: 1) - Please choose one

Suppose currentNode refers to a node in a linked list (using the Node class with member variables called data and nextNode). What statement changes currentNode so that it refers to the next node?

- ▶ currentNode ++;
- ▶ currentNode = nextNode;
- ▶ currentNode += nextNode;
- ▶ **currentNode = currentNode->nextNode;**

Question No: 6 (Marks: 1) - Please choose one

A queue where the de-queue operation depends not on FIFO, is called a priority queue

- ▶ False
- ▶ **True (Page 101)**

Question No: 7 (Marks: 1) - Please choose one

Which one is a self- referential data type?

- ▶ Stack
- ▶ Queue
- ▶ Link list
- ▶ **All of these** [Click here for detail](#)

Question No: 8 (Marks: 1) - Please choose one

Each node in doubly link list has,

- ▶ 1 pointer
- ▶ **2 pointers (Page 39)**
- ▶ 3 pointers
- ▶ 4 pointers

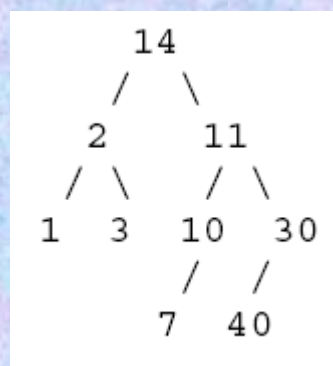
Question No: 9 (Marks: 1) - Please choose one

I have implemented the queue with a linked list, keeping track of a front pointer and a rear pointer. Which of these pointers will change during an insertion into an *EMPTY* queue?

- ▶ Neither changes
- ▶ Only front pointer changes.
- ▶ **Only rear pointer changes.**
- ▶ Both change.

Question No: 10 (Marks: 1) - Please choose one

Consider the following tree.



How many of the nodes have at least one sibling?

- ▶ 8
- ▶ 7
- ▶ 5
- ▶ 6

A sibling is an element that shares the same parent with another element

Question No: 11 (Marks: 1) - Please choose one

The nodes with no successor are called _____

- ▶ Root Nodes
- ▶ **Leaf Nodes**
- ▶ Both of these
- ▶ None of these

Question No: 12 (Marks: 1) - Please choose one

AVL Tree is,

- ▶ **Non Linear data structure** [Click here for detail](#)
- ▶ Linear data structure
- ▶ Hybrid data structure (Mixture of Linear and Non Linear)
- ▶ None of the given options.

Question No: 13 (Marks: 1) - Please choose one

We access elements in AVL Tree in,

- ▶ Linear way only
- ▶ **Non Linear way only**
- ▶ Both linear and non linear ways
- ▶ None of the given options.

Question No: 14 (Marks: 1) - Please choose one

A binary search tree should have minimum of one _____ node/s at each level,

- ▶ One
- ▶ **Two** [Click here for detail](#)
- ▶ Three
- ▶ Four

Question No: 15 (Marks: 1) - Please choose one

Consider the following statements.

- (i) A binary tree can contain at least 2^L Nodes at level L.
- (ii) A complete binary tree of depth d is a binary tree that contains 2^L Nodes at each level L between 0 and d, both inclusive.
- (iii) The total number of nodes (T_n) in a complete binary tree of depth d is $2^{d+1} - 1$.
- (iv) The height of the complete binary tree can be written as $h = \log_2 (T_n + 1) - 1$ where T_n is Total number of Nodes.

Which one of the following is correct in respect of the above statements regarding the Binary trees?

- ▶ (i) and (iii) only
- ▶ (i), (ii) and (iii) only
- ▶ (ii) and (iii) only
- ▶ **(ii), (iii) and (iv) only** [Click here for detail](#)

Question No: 16 (Marks: 1) - Please choose one
“+” is a _____ operator.

- ▶ Unary
- ▶ **Binary** (Page 64)
- ▶ Ternary
- ▶ None of the above

MIDTERM EXAMINATION

Spring 2010

Question No: 1 (M a r k s: 1)

A subscript of an array may be an integer or an integer expression.

- ▶ **True** [Click here for detail](#)
- ▶ False

Question No: 2 (M a r k s: 1)

Doubly Linked List always has one NULL pointer.

- ▶ True
- ▶ **False** (Page 43)

Question No: 3 (M a r k s: 1)

In which of the traversal method, the recursive calls can be used to traverse a binary tree ?

- ▶ **In preorder traversal only** (Page 143)
- ▶ In inorder traversal only
- ▶ In postorder traversal only
- ▶ All of the given options

Question No: 4 (M a r k s: 1)

A tree is an AVL tree if

- ▶ Any one node fulfills the AVL condition
- ▶ At least half of the nodes fulfill the AVL condition
- ▶ **All the nodes fulfill the AVL condition** (Page 213)
- ▶ None of the given options

Question No: 5 (Marks: 1)

Suppose currentNode refers to a node in a linked list (using the Node class with member variables called data and nextNode). What boolean expression will be true when cursor refers to the tail node of the list?

- ▶ **(currentNode == null)**
- ▶ (currentNode->nextNode == null)
- ▶ (nextNode.data == null)
- ▶ (currentNode.data == 0.0)

Question No: 6 (Marks: 1) - Please choose one

Suppose that the class declaration of SomeClass includes the following function prototype.

bool LessThan(SomeClass anotherObject);

Which of the following tests in the client code correctly compares two class objects alpha and beta?

- ▶ if (alpha < beta)
- ▶ **if (alpha.LessThan(beta))** [Click here for detail](#)
- ▶ if (LessThan(alpha, beta))
- ▶ if (LessThan(alpha).beta)

Question No: 7 (Marks: 1)

In C what is the operation that you can not do with primitive types?

- ▶ Assign a value to primitive type using a literal
- ▶ Declare primitive types to be constant using the Const keyword
- ▶ **Create a new instance of primitive type with New keyword** [Click here for Detail](#)
- ▶ None of these

Question No: 8 (Marks: 1)

The operation for adding an entry to a stack is traditionally called :

- ▶ add
- ▶ append
- ▶ insert
- ▶ **push** (Page 53)

Question No: 9 (Marks: 1)

The operation for removing an entry from a stack is traditionally called:

- ▶ delete
- ▶ peek
- ▶ **pop** (Page 53)
- ▶ remove

Question No: 10 (Marks: 1)

Consider the following sequence of push operations in a stack:

```
stack.push('7');  
stack.push('8');  
stack.push('9');  
stack.push('10');  
stack.push('11');  
stack.push('12');
```


- ▶ 7 8 9 10 11 12
- ▶ 9 8 11 10 7 12
- ▶ 9 10 8 11 12 7
- ▶ 9 10 8 12 7 11

Question No: 11 (Marks: 1)

_____ is the maximum number of nodes that you can have on a stack-linked list ?

- ▶ Zero
- ▶ $2n$ (where n is the number of nodes in linked list)
- ▶ **Any Number** [Click here for detail](#)
- ▶ None of these

Question No: 12 (Marks: 1)

Which of the following can be used to reverse a string value,

- ▶ **Stack** [Click here for detail](#)
- ▶ Queue
- ▶ Both of these
- ▶ None of these

Question No: 14 (Marks: 1)

AVL Tree is,

- ▶ **Non Linear data structure** [Click here for detail](#)
- ▶ Linear data structure
- ▶ Hybrid data structure (Mixture of Linear and Non Linear)
- ▶ None of the given options.

Question No: 15 (Marks: 1)

The following are statements related to queues.

- (i) The last item to be added to a queue is the first item to be removed
- (ii) A queue is a structure in which both ends are not used
- (iii) The last element hasn't to wait until all elements preceding it on the queue are removed
- (iv) A queue is said to be a last-in-first-out list or LIFO data structure.

Which of the above is/are related to normal queues?

- ▶ (iii) and (ii) only
- ▶ (i), (ii) and (iv) only
- ▶ (ii) and (iv) only
- ▶ **None of the given options** [Click here for detail](#)

Question No: 16 (Marks: 1)

An array is a group of consecutive related memory locations.

- ▶ **True** [Click here for detail](#)
- ▶ False

MIDTERM EXAMINATION

Spring 2010

Question No: 1 (Marks: 1) - Please choose one

In an array we can store data elements of different types.

- ▶ True
- ▶ **False** (Page 7)

Question No: 2 (Marks: 1) - Please choose one

In an array list the current element is

- ▶ **The first element** [Click here for detail](#)
- ▶ The middle element
- ▶ The last element
- ▶ The element where the current pointer points to

Question No: 3 (Marks: 1) - Please choose one

Which one of the following calling methods does not change the original value of the argument in the calling function?

- ▶ None of the given options
- ▶ **Call by passing the value of the argument** [Click here for detail](#)
- ▶ Call by passing reference of the argument
- ▶ Call by passing the address of the argument

Question No: 4 (Marks: 1) - Please choose one

Which one of the following statements is NOT correct?

- ▶ **Array size can be changed after its creation.** [Click here for detail](#)
- ▶ Link List size can be changed after its creation
- ▶ Binary Search Tree size can be changed after its creation
- ▶ AVL Tree size can be changed after its creation

Question No: 5 (Marks: 1) - Please choose one

Suppose that the class declaration of SomeClass includes the following function prototype. bool LessThan(SomeClass anotherObject);

Which of the following tests in the client code correctly compares two class objects alpha and beta?

- ▶ if (alpha < beta)
- ▶ **if (alpha.LessThan(beta))** [Click here for detail](#)
- ▶ if (LessThan(alpha, beta))
- ▶ if (LessThan(alpha).beta)

Question No: 6 (Marks: 1) - Please choose one

A queue is a----- data structure, whereas a stack is a -----data structure.

- ▶ **FIFO, LIFO** (Page 161,54)
- ▶ LIFO,FIFO
- ▶ none of these
- ▶ both of these

Question No: 7 (Marks: 1) - Please choose one

Which one of the following operators has higher priority than all of others?

- ▶ **Multiplication operator** [Click here for detail](#)
- ▶ Minus operator
- ▶ Plus operator
- ▶ Exponentiation operator

Question No: 8 (Marks: 1) - Please choose one

Each node in Binary Search Tree has

- ▶ 1 pointer
- ▶ **2 pointers** [Click here for detail](#)
- ▶ 3 pointers
- ▶ 4 pointers

Question No: 9 (Marks: 1) - Please choose one

Four statements about trees are below. Three of them are correct. Which one is INCORRECT?

- ▶ Trees are recursively defined multi-dimensional data structures tree
 - ▶ The order of a tree indicates a maximum number of children allowed at each node of the
 - ▶ A search tree is a special type of tree where all values (i.e. keys) are ordered
 - ▶ **If Tree1's size is greater than Tree2's size, then the height of Tree1 must also be greater than Tree2's height.**
- [Click here for detail](#)

Question No: 10 (Marks: 1) - Please choose one

Which of the following is "TRUE" about arrays,

- ▶ We can increase the size of arrays after their creation.
- ▶ We can decrease the size of arrays after their creation.
- ▶ We can increase but can't decrease the size of arrays after their creation.
- ▶ **We can neither increase nor decrease the array size after their creation.** [Click here for detail](#)

Question No: 11 (Marks: 1) - Please choose one

Searching an element in an AVL tree take maximum in AVL tree,

- ▶ $\log_2(n+1)$
time (where n is no. of nodes)
- ▶ $\log_2(n+1) - 1$
- ▶ **$1.44 \log_2 n$** (Page 227)
- ▶ $1.66 \log_2 n$

Question No: 12 (Marks: 1) - Please choose one

There is/are case/s for rotation in an AVL tree,

- ▶ 1
- ▶ 3
- ▶ 2
- ▶ **4** (Page 229)

Question No: 13 (Marks: 1) - Please choose one

Consider the following statements.

- (i) A binary tree can contain at least 2^L Nodes at level L.
- (ii) A complete binary tree of depth d is a binary tree that contains 2^L Nodes at each level L between 0 and d, both inclusive.
- (iii) The total number of nodes (T_n) in a complete binary tree of depth d is $2^{d+1} - 1$.
- (iv) The height of the complete binary tree can be written as $h = \log_2(T_n + 1) - 1$ where T_n is Total number of Nodes.

Which one of the following is correct in respect of the above statements regarding the Binary trees?

- ▶ (i) and (iii) only
- ▶ (i), (ii) and (iii) only
- ▶ (ii) and (iii) only
- ▶ (ii), (iii) and (iv) only [Click here for detail](#)

Question No: 14 (Marks: 1) - Please choose one

Consider the following infix expression.

$$5 + 6/2$$

If one converts the above expression into postfix, what would be the resultant expression?

- ▶ $56/ + 2$
- ▶ **5 6 2 / +** (Page 66)
- ▶ $5 6 / 2 +$
- ▶ $/62 + 5$

Question No: 15 (Marks: 1) - Please choose one

Which of the following is a non linear data structure?

- ▶ Linked List
- ▶ Stack
- ▶ Queue
- ▶ **Tree** (Page 112)

Question No: 16 (Marks: 1) - Please choose one

“+” is a **operator**.

- ▶ Unary
- ▶ **Binary** (Page 64)
- ▶ Ternary
- ▶ None of the above

MIDTERM EXAMINATION

Spring 2010

1. Addition of new items in stack make the pointer ----- by 2

- a. Increment, bits
- b. Increment, bytes
- c. Decrement, bits
- d. Decrement, bytes** [Click here for detail](#)

2. Next item in a linked list is known as

- a. Index
- b. Item
- c. Node** [Click here for detail](#)
- d. Child

3. What will be the postfix notation of $5+6/2$.

- a. $56+/2$
- b. $562+/$
- c. $562/+$** (Page 66)
- d. $5+62/$

4. In an AVL tree to delete a parent with two childs in a straight line following rotations will be required:-

- a. Single
- b. Double**
- c. Triple
- d. None.

5. To check the depth of an AVL tree following time will be taken:-

- a. $1.66 \log_2 n$
- b. $1.44 \log_2 n$** (Page 227)
- c. $\log_2 (n+1)-1$
- d. $1.66 \log_2 (n+1)$

6. BST is a Structure:-

- a. Linear
- b. Non Linear** [Click here for detail](#)
- c. Circular
- d. None of Above

7. After creation of an array:-

- a. Size can be increase but can not be decreased.
- b. Size can be decreased but can not be increased.
- c. Size can neither be increased nor be decreased.** [Click here for detail](#)
- d. Size can be increased and can also be decreased.

8. Each node in a BST has Pointers:-

- a. 1
- b. 2** [Click here for detail](#)
- c. 3
- d. 4

9. Highest Operators Precedence is of the following operator:-

- a. Plus
- b. Minus
- c. Multiply** [Click here for detail](#)
- d. Exponentiation

10. Following are the linear data structures:-

- a. Stacks
- b. Queues
- c. Both a & b (Page 52, 87)**
- d. None of the above

11. Each entry which points to a null value in a Singly Linked List is known as:-

- a. Node
- b. First Node
- c. Last Node**
- d. Head Node

12. Non recursive calls are faster than the Recursive calls.

- a. True (Page 323)**
- b. False

13. Tree data structure is a

- a. Linear
- b. Non Linear (Page 112) rep**
- c. Circular
- d. None of Above

14. What will be the valid postfix notation of $A+B*C-D$

- a. $ABC+*D-$
- b. $ABC*+D-$ (According to rule)**
- c. $ABCD+-*$
- d. $AB+D*C$

15. When an operator is used in between two operands this is which type of notation

- a. Prefix
- b. Postfix
- c. Infix (Page 64)**
- d. None of the Above

MIDTERM EXAMINATION

Spring 2009

Question No: 1 (Marks: 1) - Please choose one

Which one of the following is a valid postfix expression?

- ▶ $ab+c*d-$
- ▶ **$abc*+d-$ (According to rule)**
- ▶ $abc+*d-$
- ▶ $(abc*)+d-$

Question No: 2 (Marks: 1) - Please choose one

The tree data structure is a

- ▶ Linear data structure
- ▶ **Non-linear data structure (Page 112)**
- ▶ Graphical data structure
- ▶ Data structure like queue

Question No: 3 (Marks: 1) - Please choose one

A **Compound Data Structure** is the data structure which can have multiple data items of same type or of different types. Which of the following can be considered compound data structure?

- ▶ **Arrays** [Click here for detail](#)
- ▶ LinkLists
- ▶ Binary Search Trees
- ▶ All of the given options

Question No: 4 (Marks: 1) - Please choose one

Suppose a pointer has been declared in main but has not assigned any variable address then

- ▶ That pointer points to First byte in main function
- ▶ **That pointer contains a NULL value**
- ▶ None of these
- ▶ **That pointer points to any memory address**

Question No: 5 (Marks: 1) - Please choose one

Here is the start of a C++ class declaration:

```
class foo
{
public:
void x(foo f);
void y(const foo f);
void z(foo f) const;
...
```

Which of the three member functions can alter the PRIVATE member variables of the foo object that activates the function?

- ▶ **Only x can alter the private member variables of the object that activates the function.**
- ▶ Only y can alter the private member variables of the object that activates the function.
- ▶ Only z can alter the private member variables of the object that activates the function.

► **Two of the functions can alter the private member variables of the object that activates the function.**

Only the x and y can alter the private member variable of the foo class object. Last Option is more correct but not exact. In the last option the two function name are not mentioned

Question No: 6 (Marks: 1) - Please choose one

The operation for removing an entry from a stack is traditionally called:

- delete
- peek
- **pop (Page 53)**
- remove

Question No: 7 (Marks: 1) - Please choose one

Which statement of the following statements is incorrect?

- Lists can be implemented by using arrays or linked lists
- A list is a sequence of one or more data items
- Stack is a special kind of list in which all insertions and deletions take place at one end
- **Stacks are easier to implement than lists**

Question No: 8 (Marks: 1) - Please choose one

Parameters in function call are passed using,

- **Stack (Page 80)**
- Queue
- Binary Search Tree
- AVL Tree

Question No: 9 (Marks: 1) - Please choose one

Consider the following sequence of push operations in a stack:

```
stack.push('7');  
stack.push('8');  
stack.push('9');  
stack.push('10');  
stack.push('11');  
stack.push('12');
```

- **7 8 9 10 11 12**
- 9 8 11 10 7 12
- 9 10 8 11 12 7
- 9 10 8 12 7 11

Question No: 10 (Marks: 1) - Please choose one

What is the maximum depth of recursive calls a function may make?

- 1
- 2
- n (where n is the argument)
- **There is no fixed maximum**

Question No: 11 (Marks: 1) - Please choose one

Consider the following function:

```
void test_a(int n)
{
    cout << n << " ";
    if (n>0)
        test_a(n-2);
}
```

What is printed by the call test_a(4)?

- ▶ **4 2 0**
- ▶ 0 2 4
- ▶ 0 2
- ▶ 2 4

Question No: 12 (Marks: 1) - Please choose one

Queue follows,

- ▶ Last in First out
- ▶ First in Last out
- ▶ **First in First out** (Page 87)
- ▶ None of these

Question No: 13 (Marks: 1) - Please choose one

is a binary tree where every node has a value, every node's left subtree contains only values less than or equal to the node's value, and every node's right subtree contains only values that are greater than or equal ?

- ▶ **Strictly Binary Tree**
- ▶ **Binary Search tree** [Click here for detail](#)
- ▶ AVL tree
- ▶ All of these

Question No: 14 (Marks: 1) - Please choose one

Four statements about trees are below. Three of them are correct. Which one is INCORRECT?

- ▶ **Trees are recursively defined multi-dimensional data structures** [Click here for detail](#)
- ▶ The order of a tree indicates a maximum number of children allowed at each node of the tree
- ▶ A search tree is a special type of tree where all values (i.e. keys) are ordered
- ▶ If Tree1's size is greater than Tree2's size, then the height of Tree1 must also be greater than Tree2's height.

Question No: 15 (Marks: 1) - Please choose one

Below is a binary search tree. If we delete the value 50 using the algorithm we discussed, what value will be in the root of the remaining tree?

- ▶ 50
- ▶ 60
- ▶ 70
- ▶ **80**

Question No: 16 (Marks: 1) - Please choose one

Is a data structure that can grow easily dynamically at run time without having to copy existing elements?

- ▶ Array
- ▶ **List**
- ▶ Both of these
- ▶ None of these

MIDTERM EXAMINATION

Spring 2009

Question No: 1 (Marks: 1) - Please choose one

Which one of the following statement is NOT correct .

- ▶ **In linked list the elements are necessarily to be contiguous** [Click here for detail](#)
- ▶ In linked list the elements may locate at far positions in the memory
- ▶ In linked list each element also has the address of the element next to it
- ▶ In an array the elements are contiguous

Question No: 2 (Marks: 1) - Please choose one

In a program a reference variable, say x, can be declared as

- ▶ **int &x ;** [Click here for detail](#)
- ▶ int *x ;
- ▶ int x ;
- ▶ None of the given options

Question No: 3 (Marks: 1) - Please choose one

Linked lists are collections of data items "lined up in a row" , insertions and deletions can be made only at the front and the back of a linked list.

- ▶ True
- ▶ **False** [Click here for detail](#)

Question No: 4 (Marks: 1) - Please choose one

A Linear Data Structure is the data structure in which data elements are arranged in a sequence or a linear list. Which of the following is Non Linear Data Structure?

- ▶ Arrays
- ▶ LinkLists
- ▶ **Binary Search Trees** [Click here for detail](#)
- ▶ None of these

Question No: 5 (Marks: 1) - Please choose one

A queue where the de-queue operation depends not on FIFO, is called a priority queue

- ▶ False
- ▶ **True** (Page 101)

Question No: 6 (Marks: 1) - Please choose one

Which one of the following statements is correct?

- ▶ **Array size is fixed once it is created** [Click here for detail](#)
- ▶ Link List size is fixed once it is created.
- ▶ Binary Search Tree size is fixed once it is created
- ▶ AVL Tree size is fixed once it is created

Question No: 7 (Marks: 1) - Please choose one

Which one of the following is correct about pointers?

- ▶ They always point to different memory locations
- ▶ They may point to a single memory location
- ▶ The address of two pointer variables is same
- ▶ None of these

Question No: 8 (Marks: 1) - Please choose one

Which of the following abstract data types are NOT used by Integer Abstract Data type group?

- ▶ short
- ▶ Int
- ▶ **float** [Click here for detail](#)
- ▶ long

Question No: 9 (Marks: 1) - Please choose one

The operation for adding an entry to a stack is traditionally called :

- ▶ add
- ▶ append
- ▶ insert
- ▶ **push** (Page 53)

Question No: 10 (Marks: 1) - Please choose one

The operation for removing an entry from a stack is traditionally called:

- ▶ delete
- ▶ peek
- ▶ **pop** (Page 53)
- ▶ remove

Question No: 11 (Marks: 1) - Please choose one

We can add elements in QUEUE From _____

- ▶ Front
- ▶ **Rear** (Page 91)
- ▶ From Both Rare and Front
- ▶ None of these

Question No: 12 (Marks: 1) - Please choose one

The difference between a binary tree and a binary search tree is that ,a binary search tree has

▶ **two children per node whereas a binary tree can have none, one, or two children per node** [Click here for detail](#)

- ▶ in binary search tree nodes are inserted based on the values they contain
- ▶ in binary tree nodes are inserted based on the values they contain
- ▶ none of these

Question No: 13 (Marks: 1) - Please choose one

Suppose n is the number of nodes in a complete Binary Tree then maximum steps required for a search operation are,

▶ **$\log_2(n+1) - 1$** (Page 139)

- ▶ $\log_2(n+1)$
- ▶ $\log_2(n) - 1$
- ▶ $\log_2(n)$

Question No: 14 (Marks: 1) - Please choose one

The following is a segment of a C program.

```
int pqr(BinaryNode t)
{ if (t == null )
  return -1;
else
  return 1+max(pqr(t.left),pqr(t.right)) }
```

Identify, what the above program intend(s) to do?

- ▶ Compute the height of a binary tree using an in-order traversal
- ▶ Compute the height of a binary tree using a pre-order traversal
- ▶ Compute the depth of a binary tree using a pre-order traversal
- ▶ Compute the depth of a binary tree using a post-order traversal

Question No: 15 (Marks: 1) - Please choose one

Consider the following infix expression:

$3 + 5 * 6 - 7 * (8 + 5)$

Which of the following is a correct equivalent expression(s) for the above?

- ▶ $3 \ 6 \ 5 + * \ 7 \ 5 \ 8 + - *$
- ▶ $3 \ 6 \ 5 \ 7 \ 5 \ 8 + * + - *$
- ▶ $3 \ 5 \ 6 + * \ 7 \ 8 \ 5 + - *$
- ▶ **$3 \ 5 \ 6 * + \ 7 \ 8 \ 5 + * -$**

Question No: 16 (Marks: 1) - Please choose one

An array is a group of consecutive related memory locations.

- ▶ **True** [Click here for detail](#)
- ▶ False

Question No: 17 (Marks: 1)

Muhammad Moaaz Siddiq MCS (2nd)
mc100401285@gmail.com
Campus:- Institute of E-Learning &
Modern Studies (IEMS) Samundar
(PSMD01)

Muhammad Irfan Farid
MCS(2nd)
virtualcheater@gmail.com
Virtual University Campus . D.G. Khan
(VDGK01)

Muhammad Awais- MCS(2nd)
awais4you@gmail.com
Campus : -Abbas Institute Of
Modern Studies(AIMS)Lahore
(PLHR25)

Is this a correct statement? Give answer in Yes or No.

A node cannot be deleted, when the node to be deleted has both left and right subtrees.

False ---- No, it can be deleted.

Question No: 18 (Marks: 1)

Deleting a leaf node in binary search tree involves setting _____ pointer/s of that node's parent as null.

- 1
- 2
- 3
- 4

CS 301 150+ MCQs Solved with References

A template provides a convenient way to make a family of.

variables and data members

functions and classes

classes and exceptions

programs and algorithms

A class template may inherit from another class template.

True

False

Target of a _____ function call is determined at run time.

instance

virtual

operator

none of given

A class hierarchy .

shows the same relationships as an organization chart.

describes “has a” relationships.

describes “is a kind of” relationships.

shows the same relationships as a family tree.

Sender of the message does not need to know the exact class of receiver in_____.

Abstraction

Polymorphism

Inheritance

none of the given

A function call is resolved at run-time in_____ .

non-virtual member function

virtual member function

Both non-virtual member and virtual member function.

None of given

Adding a derived class to a base class requires fundamental changes to the base class.

True

False

User can make virtual table explicitly.

True

False

Binding means that target function for a call is selected at compile time.

Static

Dynamic

Automatic

None of given

Target of a _____ function call is determined at run time.

instance

virtual

operator

none of given

Which line will produce error. Class phone: private Transmit, private Receiver { }

1. int main() 2. { 3. phone obj; 4. Transmit* obj1 = &obj; 5. Received obj2 = &obj; 6. }.

3rd line will produce error

4th line will produce error

3rd and 4th line will produce error.

5th line will produce error

Function overriding is done in context of,

Single class

Single derived class

Single base class

Derived and base classes

Consider the code below, class class1{ public: void func1(); }; class class2 : protected class1 { }; Function func1 of class1 is _____ in class2,

public

protected

private

none of the given options

The following statements: 1) int iArray[5]; 2) int *pArr = iArray;

These statements will compile successfully

Error in first statement

Error in second statement

None of given options

Methodologies to the development of reusable software relate to_____.

Structure programming

procedural programming

generic programming

None of the given

Function template must have a parameter.

True

False

The default inheritance mode is,

- Public inheritance
- Protected inheritance
- Private inheritance**
- None of these options

Two functions with same names, parameters and return type can exist in,

Function overloading

- Function overriding
- Operator overloading
- None of these options

Consider the code below, class c1{ }; class c2 : public c1 { }; class c3 : public c2 { }; Then c2 is,

Direct base class of c3

- Direct child class of c3
- Direct base class of c1
- None of these

Virtual functions allow you to

- create an array of type pointer-to-base class that can hold pointers to derived classes.
- create functions that can never be accessed.

group objects of different classes so they can all be accessed by the same function code.

- use the same function call to execute member functions of objects from different classes.

User can make virtual table explicitly.

- True
- False**

In order to define a class template the first line of definition must be:

template <typename T>

typename <template T>

Template Class <ClassName>

Class <Template T>

Consider the following statements: 1) int iArray[5]; 2) int *pArr = iArray;

These statements will compile successfully

Error in first statement

Error in second statement

None of given options

In c++ dynamic binding and polymorphism will be achieved when member function will be ____.

- private
- public**

virtual

inline

In type in depended function template should be use where code and behavior must be identical.

True

False

Consider the code below, class class1{ protected: int i; }; class class2 : private class1 { }; Then int member i of class1 is ____ in class2,

public

protected

private

none of the given options

In specialization we can,

Replace child class with its base class

Replace base class with its child class (Not Sure)

Replace both child and base classes interchangeably

None of the given options

Consider the code below, class class1{ public: void func1(); }; class class2 : public class1 { }; Function func1 of class1 is ____ in class2,

public

protected

private

none of the given options

It is illegal to make objects of one class members of another class.

True

False

An abstract class is useful when

no classes should be derived from it.

there are multiple paths from one derived class to another.

no objects should be instantiated from its.

you want to defer the declaration of the class.

In resolution order compiler search firstly ____.

Generic Template

Partial Specialization

Complete Specialization

Ordinary function

template<> class Vector{ void p; //.... void*& operator[] ((int i); };**

This specialization can then be used as the common implementation for all Vectors of pointers.

This specialization can then be used as the all type implementation for one type classes.

This specialization can then be used double type pointers.

This specialization should be used for Vectors of all type int types.

In private inheritance derived class pointer can be assigned to base class pointer in.

Main function

In derived class member and friend functions

In base class member and friend functions

None of the given options

Which statement will be true for concrete class?

it implements an virtual concept.

it can be instantiated

it cannot be instantiated

none of given

Target of a _____ function call is determined at run time.

instance

virtual

operator

none of given

The Specialization pattern after the name says that this specialization is to be used for every_____.

data types

meta types

virtual types

pointers type

c++ dynamic binding and polymorphism will be achieved when member function will be ____.

private

public

virtual

inline

Consider the code below, class class1{ protected: void func1(); }; class class2 : public class1 { }; Function func1 of class1 is _____ in class2,

public

protected

private

none of the given options

Consider the code below, class class1{ protected: int i; }; class class2 : protected class1 { }; Then int member i of class1 is _____ in class2,

public

protected

private

none of the given options

Consider the code below, class class1{ private: void func1(); }; class class2 : private class1 { }; Function func1 of class1 is ____ in class2,

public

protected

private

none of the given options

Target of a ____ function call is determined at run time.

instance

virtual

operator

none of given

Consider the following statements: 1) int iArray[5]; 2) int *pArr = iArray;

These statements will compile successfully

Error in first statement

Error in second statement

None of given options

Consider the code below, class class1{ private: int i; }; class class2 : private class1 { }; Then int member i of class1 is ____ in class2,

public

protected

private

none of the given options

If there is a pointer, p, to objects of a base class, and it contains the address of an object of a derived class, and both classes contain a virtual member function, ding(), then the statement p->ding(); will cause the version of ding() in the ____ class to be executed.

base

derived

virtual

implemented

A class template may inherit from another class template.

True

False

Derived class can inherit from public base class as well as private and protected base classes

True

False

Two functions with same names, parameters and return type can exist in, Function overloading

Function overriding
Operator overloading
None of these options

Consider the code below, class class1{ private: int i; }; class class2 : public class1 { }; Then int member i of class1 is ____ in class2,
public
protected
private
none of the given options

Target of a ____ function call is determined at run time.
instance
virtual
operator
none of given

A class template may inherit from another class template.
True
False

A function call is resolved at run-time in ____
non-virtual member function
virtual member function

Both non-virtual member and virtual member function.
None of given

A class hierarchy
shows the same relationships as an organization chart.
describes “has a” relationships.
describes “is a kind of” relationships.
shows the same relationships as a family tree.

Consider the code below, class class1{ public: int i; }; class class2 : public class1 { }; Then int member i of class1 is ____ in class2,
public
protected
private
none of the given options

Consider the code below, class c1{ }; class c2 : public c1 { }; class c3 : public c2 { }; Then c1 is,
Direct base class of c3
Direct child class of c3
Direct base class of c2
Direct child class of c2

A class can inherit from more than one class is called.
Simple inheritance

Multiple inheritances

Single inheritance

Double inheritance

template<T> class Vector{ void p; //.... void*& operator[] ((int i); };**

This specialization can then be used as the common implementation for all Vectors of pointers.

This specialization can then be used as the all type implementation for one type classes.

This specialization can then be used double type pointers.

This specialization should be used for Vectors of all type int types.

Consider the code below, class class1{ public: int i; }; class class2 : protected class1 { }; Then int member i of class1 is _____ in class2,

public

protected

private

none of the given options

Consider the code below, class class1{ private: void func1(); }; class class2 : public class1 { }; Function func1 of class1 is _____ in class2,

public

protected

private

none of the given options

Templates automatically create different versions of a function, depending on user input.

True

False

_____ **Binding means that target function for a call is selected at run time**

Automatic

Dynamic

Static

Dramatic

When we create objects, then space is allocated to:

Member function

Access specifier

Data member

None of given

There is only one form of copy constructor.

True

False

Which of the following features of OOP is used to deal with only relevant details?

Abstraction

Information hiding Object

_____ Binding means that targets function for a call is selected at compile time.

Static

Dynamic

Automatic

None of given

A Class hierarchy

Shows the same relationships as an organization chart

Describes “has a” relationships.

Describes “is a kind of” relationships.

Shows the same relationships as a family tree

In C++, we declare a function virtual by preceding the function header with keyword “Inline”.

True

False

It is illegal to make objects of one class members of another class.

True

False

In Resolution order compiler search firstly_____.

Generic Template

Partial Specification

Complete Specification

Ordinary function

Derived class can inherit from public base class as well as private and protected base classes

True

False

**Which line will produce error. Class phone: Private Transmit, private Receiver { }
1.int main () 2. { 3.phone obj; 4.Tranmit*obj1 = &obj; 5.Received obj2 = &obj; 6.}**

3rd line will produce error

4th line will produce error

3rd and 4th line will produce error.

5th line will produce error.

Methodologies to the development of reusable software relate to _____.

Structure programming

Procedural programming

Generic programming

None of the given

A template argument is preceded by the keyword _____.

Vector

Class

Template

Type*

Friends are used exactly the same for template and non-template classes.

True

False

A function template must have a parameter

True

False

Child class can call constructor of its,

Direct base class

Indirect base class

Both direct and indirect base classes

None of these.

Which statement will be true for concrete class?

It implements an virtual concept.

It can be instantiated

It cannot be instantiated

None of given

A class D can be derived from a class C, which is derived from a class B, which is derived from a class A

True

False

Adding a derived class to a base class requires fundamental changes to the base class.

True

False

A Class or class template can have member _____ that are themselves templates.

Variable

Function

Objects

None of given

Which will be the Primary task or tasks of generic programming?

- Categorize the abstractions in a domain into concepts
- Implement generic algorithms based on the concepts
- Build concrete models of the concepts
- All of given**

The default inheritance mode is,**Public inheritance**

- Protected Inheritance
- Private Inheritance
- None of these options

If there is a pointer, p, to objects of a base class, and it contains the address of an object of a derived class, and both classes contain a virtual member function, ding(), then the statement p->ding(); will cause the version of ding() in the _____ class to be executed.

- Base
- Derived
- Virtual**
- Implemented

Sender of the message does not need to know the exact class of receiver in_____.

- Abstraction
- Polymorphism
- Inheritance
- none of the given

Question # 1 of 10

Information hiding can be achieved through_____.

1. Encapsulation, Inheritance
2. Encapsulation, Polymorphism
3. **Encapsulation, Abstraction**
4. Overloading

Question # 2 of 10 (Start time: 01:11:21 AM) Total M - 1

A good model is related to a real life problem.

Select correct option:

1. Loosely
2. Openly

3. **Closely**

Question # 3 of 10 (Start time: 01:12:33 AM) Total M - 1

Which of the following features of OOP is used to derive a class from another?

Select correct option:

1. Encapsulation
2. Polymorphism
3. Data hiding
4. **Inheritance**

Question # 4 of 10 (Start time: 01:13:51 AM) Total M - 1

Which of the following is a weak relationship between two objects?

Select correct option:

1. Inheritance
2. Composition
3. **Aggregation**
4. None of given

Question # 5 of 10 (Start time: 01:14:56 AM) Total M - 1

Data items in a class must be private.

Select correct option:

1. True
2. **False**

Question # 6 of 10 (Start time: 01:15:52 AM) Total M - 1

Which one is a class association

Select correct option:

1. Simple Association
2. **Inheritance**
3. Composition
4. Aggregation

Question # 7 of 10 (Start time: 01:16:55 AM) Total M - 1

Suppose there is an object of type Person, which of the following can be considered as one of its attributes

Select correct option:

1. Name
2. Age
3. Work()
4. **Both Name and Age**

Question # 8 of 10 (Start time: 01:17:52 AM) Total M - 1

Which one is not an object association?

Select correct option:

1. Simple association
2. **Inheritance**
3. Aggregation
4. Association

Question # 9 of 10 (Start time: 01:18:50 AM) Total M - 1

Using encapsulation we can achieve

Select correct option:

1. **Information hiding**
2. Least interdependencies among modules
3. Implementation independence
4. All of given options

Question # 10 of 10 (Start time: 01:19:43 AM) Total M - 1

In constant member function the type of this pointer is:

Select correct option:

1. Constant pointer
2. **Constant pointer to object**
3. Constant pointer to class
4. Constant pointer to constant object

Question # 1 of 10

Which of the following is the way to extract common behavior and attributes from the given classes and make a separate class of those common behaviors and attributes?

1. **Generalization**
2. Sub-typing
3. Specialization
4. Extension

Question # 2 of 10

The ability to derive a class from more than one class is called

1. Single inheritance
2. Encapsulation
3. **Multiple inheritance**
4. Polymorphism

Question # 3 of 10:

If MyClass has a destructor what is the destructor named?

1. MyClass
2. **~MyClass**
3. My~Class
4. MyClass~

Question # 4 of 10:

Class abc{ ----- }; Is a valid class declaration?

1. **yes**
2. no

Question # 5 of 10:

Without using Deep copy constructor, A _____ problem can occur

1. System crash
2. Memory Leakage
3. Dangling pointer
4. **All of the given**

Question # 6 of 10:

If only one behaviour of a derived class is incompatible with base class, then it is:

1. Generalization
2. **Specialization**
3. Extension
4. Inheritance

Question # 8 of 10:

Only tangible things can be chosen as an object.

- True
- **False**

When we create objects, then space is allocated to:

Member functions

Access specifier

Data members

None of the given

If we extend our model, and the rest of model is not affected, then its called

Consistency

Flexibility

Efficiency

Reusability

Using encapsulation we can achieve

Information hiding

Least interdependencies among modules

Implementation independence

All of given options

Consider the statement “room has chair” which of the following type of association exists between room and chair?

Inheritance

Composition

There is no association

Aggregation

A good model is related to a real life problem.

Loosely

Openly

Closely

Not

If a class A inherits from class B, then class A is called.

Child Class

Derived class

Parent class

Child and derived class

Which one of the following is not a major element in an object model?

Abstraction

Encapsulation

Persistence

Hierarchy

Member functions defined inside a class declaration are _____ by default.

Visible

Public

Private
Changed

Advantage(s) of information hiding
Simplifies the model

Restriction to change

Both of above
None of the above

Which of the following parts of an object exhibits its state?

Data
Operations
Any public part
Any private part

Without using Deep copy constructor. A _____ problem can occur

System crash
Memory
Leakage
Dangling pointer
All the given

Three main characteristics of "Object Oriented programming" are,

Encapsulation, dynamic binding, polymorphism

Polymorphism, overloading, overriding

Encapsulation, inheritance, dynamic , binding

Encapsulation, inheritance, polymorphism

Overriding concept is used in"

Encapsulation
Polymorphism
Information hiding
None of the given

Main advantage of inheritance is:

Better understanding

Less understating
Reuse
Less complexity

Which one is a class association

Simple association

Inheritance

Composition

Aggregation

Public member function exposed by a class is called surface

True

False

Constructor and destructor can be declared constant

True

False

Which can allocate dynamic memory allocation through_____ keyword.

Static

Global

New

Const

In inheritance, a child class is sub-type of base class.

True

False

If only one behaviour of a derived class is incompatible with base class, then it is:

Generalization

Specification

Extension

Inheritance

“A fan has wings”. Which type of relation exists between fan and wings in the sentence?

Aggregation

Association

Generalization

Composition

When an object initialize _____ automatically call by the complier

Constructor

Function

Object

None of the given

A class _____ destructor.

No

One

Two

Three

Suppose there is an object of type person, which of the following can be considered as one of its attributes

Name

Age

Work ()

Both name and Age

If MyClass has destructor named?

MyClass

~MyClass

My~Class

MyClass~

An abstract class shows _____ behaviour.

Overriding

Specific

General

None of the given

The _____ keyword tells the compiler to substitute the code within the function definition for every instance of a function call.

Virtual

Inline

Instance

None of the given

We can choose any arbitrary name for a constructor.

Select correct option:

True

False

Which one is not the main feature of object oriented programming?

Select correct option:

Classes and Objects

Exception handling

Inheritance

Polymorphism

The dot operator (or class member access operator) connects the following two entities (reading from left to right):

Select correct option:

A class member and a class object

A class object and a class

A class and a member of that class

A class object and a member of that class

If only one behaviour of a derived class is incompatible with base class, then it is:

Select correct option:

Generalization

Specialization

Extension

Inheritance

Storing data and functions in a single unit (class). Data cannot be accessible to the outside world and only those functions which are stored in the class can access it

Select correct option:

Inheritance

Abstraction

Polymorphism

Encapsulation

If MyClass has a destructor what is the destructor named?

Select correct option:

MyClass

~MyClass

My~Class

MyClass~

A real world object can be transformed into programming entity by defining its respective

Select correct option:

Class

Function

Only states

Only behaviour

Which of the following features of OOP is used to deal with only relevant details?

Select correct option:

Abstraction

Information hiding

Object

Inheritance

In class, attributes and behaviour is represented by:

Select correct option:

Member functions, data members

Member functions, scope of data members

Data members, member functions

None of the given

Using encapsulation we can achieve

Select correct option:

Information hiding

Least interdependencies among modules

Implementation independence

All of given options

Information hiding can be achieved through_____.

Encapsulation, Inheritance

Encapsulation, Polymorphism

Encapsulation, Abstraction

Overloading

A good model is related to a real life problem.

Select correct option:

Loosely

Openly
Closely

Which of the following features of OOP is used to derive a class from another?

Select correct option:

Encapsulation

Polymorphism

Data hiding

Inheritance

Which of the following is a weak relationship between two objects?

Select correct option:

Inheritance

Composition

Aggregation

None of given

Data items in a class must be private.

Select correct option:

True

False

Which one is a class association?

Select correct option:

Simple Association

Inheritance

Composition

Aggregation

Suppose there is an object of type Person, which of the following can be considered as one of its attributes

Select correct option:

Name

Age

Work()

Both Name and Age

Question # 8 of 10 (Start time: 01:17:52 AM) Total M a r k s: 1

Which one is not an object association?

Select correct option:

Simple association

Inheritance

Aggregation

Association

Using encapsulation we can achieve

Select correct option:

Information hiding

Least interdependencies among modules

Implementation independence

All of given options

In constant member function the type of this pointer is:

Select correct option:

Constant pointer

Constant pointer to object

Constant pointer to class

Constant pointer to constant object

Which of the following is the way to extract common behavior and attributes from the given classes and make a separate class of those common behaviors and attributes?

Generalization

Sub-typing

Specialization

Extension

The ability to derive a class from more than one class is called

Single inheritance

Encapsulation

Multiple inheritance

Polymorphism

If MyClass has a destructor what is the destructor named?

MyClass

~MyClass

My~Class

MyClass~

Class abc{ ----- }; Is a valid class declaration?

yes

no

Without using Deep copy constructor, A _____ problem can occur

System crash

Memory Leakage

Dangling pointer

All of the given

If only one behaviour of a derived class is incompatible with base class, then it is:

Generalization

Specialization

Extension

Inheritance

Which of the following may not be an integral part of an object?

state

behavior

Protected data members

All of given

Only tangible things can be chosen as an object.

True

False

Class is not a mechanism to create objects and define user data types.

true

false

Memory is allocated to non static members only, when:

Class is created

Object is defined

Object is initialized

Object is created

The sub-object's life is not dependent on the life of master class in _____.

Composition

Aggregation

Separation

non of the given

Unary operators and assignment operator are right associative.

true

false

The >= operator can't be overloaded.

true

false

_____ is creating objects of one class inside another class.

Association

Composition

Aggregation

Inheritance

If we are create array of objects through new operator, then

We can call overloaded constructor through new

We can't call overloaded constructor through new

We can call default constructor through new

None of the given

Object can be declared constant with the use of Constant keyword.

true

false

_____ Operator will take only one operand.

New

int

object

none of the given

Which of the following operator(s) take(s) one or no argument if overloaded?

++

*

%

All of the given choices

this pointer does not pass implicitly to _____ functions.

Static Member

Non-Static Member

Instance Number
None of the given

Operator overloading is
making C++ operators work with objects.
giving C++ operators more than they can handle.
giving new meanings to existing Class members.
making new C++ operators

Which of the following is TRUE,
Derived class pointer can be used as Base class pointer
Base class pointer can be used as Derived class pointer
Both of these options
None of these options

_____ Binding means that target function for a call is selected at run time
Automatic
Dynamic
Static
Dramatic

When we want to implement one class in terms of another class then we use
Public inheritance
Protected inheritance
Private inheritance
None of these options

Consider the code below, class c1{ }; class c2 : public c1 { }; class c3 : public c2 { };
Then c2 is
Direct base class of c3
Direct child class of c3
Direct base class of c1
None of these

Consider the code below, class class1{ private: int i; }; class class2 : public class1 { };
Then int member i of class1 is _____ in class2,
Public
Protected
Private

None of the given options

A parent class can call constructor of its child class through

Its constructor initialization list

Its constructor body

Both from its constructor initialization list or body

Can not call the constructor of its child class

In Private Inheritance the public members of base class become _____ in derived class.

Public

Private

Protected

None of the given options

Function overriding is done in context of,

Single class

Single derived class

Single base class

Derived and base classes

Child class can call constructor of its,

Direct base class

Indirect base class

Both direct and indirect base classes

None of these

Adding a derived class to a base class requires fundamental changes to the base class

True

False

_____ is automatically called when the object is created.

Member function

Object

Constructor

None of the given

Inheritance is a way to

Organize data

Pass arguments to objects of classes

Add features to existing classes without rewriting them

Improved data-hiding and encapsulation

Through interface we access object_____.

States

Data members

Behaviour

None of the given

Which of the following may not be integral part of an object?

State

Behavior

Protected data members

All of given

Suppose there is an object of type Person, which of the following can be considered as one of its attributes.

Name

Age

Work()

Both Name and Age

If we have an overloaded constructor in our class, then compiler constructs the default constructor.

True

False

Advantages(s) of information hiding

Simplifies the model

Restriction to change

Both of above

None of the above

If some of objects exhibit identical characteristics, then they belong to:

Different classes

Multiple classes

Same class

None of the given

Which of the following is the way to extract common behaviour and attributes from the given and make a separate class of those common behaviours and attributes?

Generalization

Sub-typing

Specification

Extension

A class has _____ destructor

One

Two

Three

Behaviors of a type of objects are represented through

States

Attributes

Operations

Which is not an example of multiple inheritances?

Mermaid

Amphibious Vehicle

Car

None of the above

Static data members are called _____ variable

Class

Object

Structure

None of the given

Associatively can be change in operator overloading.

True

False

_____ and _____ methods may not be declared abstract.

Private,static

private,public

static,public

none of the given

Let Suppose a class Student with objects std1, std2, and std3. For the statement std3 = std1 - std2 to work correctly, if the overloaded - operator must

take two arguments.

None of the given choices

take single argument

take three arguments

To initialize an array of objects, only _____ will be called

Default Constructor

Overloaded Constructor

Default Object

None of the above

_____ provide the facility to access the data member.

accesser function

private function

inline function

None of the given

If a class involves dynamic memory allocation, then:

Default copy constructor, shallow copy is implemented

User defined copy constructor, shallow copy is implemented

Default copy constructor, deep copy is implemented

User defined copy constructor, deep copy is implemented

The sentence "Object Oriented Programming book in bookshelf" is an example of:

Association

Multiple associations

Aggregation

How the information hidden within an object can be accessed?

Through its interface

Through its private data member

Through its private member functions

There is no need to is or is a kind of relationship in generalization

True

False

An object has _____ interface(s)

Only one

Only two

One or more than one

Which of the following is an advantage of OOP?

It provides an ability to create one user defined data type extending the other

It provides the facility of defining abstract data type through which real entities can be defined better

All of the given option

Encapsulation means

Select correct option:

Extending the behaviour of class in another class

Data and behaviour are tightly coupled within an entity

One entity takes all the attributes and operations of the other

Taking out the common features and put those in a separate class

We can allocate dynamic memory allocation through _____ key word.

Select correct option:

Static

Global

New

Const

In programming, where the actual actions are coded is called:

Select correct option:

Function declaration

Function calling

Function definition

None of the given

Which of the following is directly related to polymorphism?

Select correct option:

Overriding

Const members

Static members

None of given

In a class declaration, data or functions designated private are accessible

Select correct option:

to any function in the program.

only if you know the password.

to member functions of that class.

only to public members of the class.

An instance of user defined type is called

Select correct option:

Object

Class

both of above

none of above

Aggregation is implemented using pointer.

Select correct option:

True

False

If class A supports all the operations of class B, then class A is:

Class A is behaviourally compatible with class B

Class A is behaviourally incompatible with class B

Class A is independent of class B

None of the given

Objects having identical characteristics belong to _____

Same class

Two different classes

Any number of different classes

Objects can not have identical characteristics



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CS301 Midterm Solved McQs Papers By Waqar Sidhu (File 1)

Question No 1

Each Node in singly link list,

Select Correct Option

✓ **1 pointer**.....(For More Visit VUStudents.pk)

✓ 2 pointers

✓ 3 pointers

✓ 4 pointers

Question NO 2

Parameters in function call are passed using,

Select Correct Option

✓ **Stack**.....(For More Visit VUStudents.pk)

✓ Queue

✓ Binary Search Tree

✓ AVL Tree

Question NO 3

The _____ method of list data structure removes the element residing at the current position.

Select Correct Option

✓ Add

✓ Next

✓ **Remove**.....(For More Visit VUStudents.pk)

✓ Find

Question NO 4

Insertion in a linked list can be done at

Select Correct Option

✓ Front only

✓ Back only

✓ Somewhere in middle only

✓ **Front, back and somewhere in the middle**.....(For More Visit VUStudents.pk)

Question NO 5

An array is a group of ----- memory locations.

Select Correct Option

✓ Scattered

✓ Isolated

✓ Random(non-consecutive)

✓ **Consecutive**.....(For More Visit VUStudents.pk)



For More Visit VUStudents.pk





Question NO 6

Which of the following applications may use a stack?

Select Correct Option

- ✓ Accessing shared resources
- ✓ **Parentheses balancing program**.....(For More Visit VUStudents.pk)
- ✓ Buffering messages
- ✓ Waiting list

Question NO 7

Every AVL is _____.

Select Correct Option

- ✓ Ternary Tree
- ✓ Complete Binary Tree
- ✓ Heap
- ✓ **Binary Search Tree**.....(For More Visit VUStudents.pk)

Question NO 8

Consider the following infix expression:

$$x - y * a + b / c$$

Which of the following is a correct equivalent expression for the above?

Select Correct Option

- ✓ $x y - a * b + c /$
- ✓ $x * y a - b c / +$
- ✓ **$x y a * - b c / +$**(For More Visit VUStudents.pk)
- ✓ $x y a * - b / + c$

Question NO 9

Non-recursive calls are faster than which of the following calls?

Select Correct Option

- ✓ Parameterized
- ✓ **Recursive**.....(For More Visit VUStudents.pk)
- ✓ Function
- ✓ Non-Function

Question NO 10

The balance of a node in the result of “height of left subtree” “height of right subtree.”

Select Correct Option

- ✓ Plus
- ✓ **Minus**.....(For More Visit VUStudents.pk)
- ✓ Multiply
- ✓ Divided by





Question NO 11

Searching an element in an AVL tree takes maximum _____ time (when n is number of nodes in AVL tree)

Select Correct Option

- ✓ $\log_2(n+1)$
- ✓ $\log_2(n+1)-1$
- ✓ **$1.44 \log_2 n$ (For More Visit VUStudents.pk)**
- ✓ $1.66 \log_2 n$

Question NO 12

A complete binary tree having “N” nodes consists of Levels.

Select Correct Option

- ✓ **$\log_2(N+1) - 1$ (For More Visit VUStudents.pk)**
- ✓ $\log_2(N-1) - 1$
- ✓ $\log_2(N+1) + 1$
- ✓ $\log_2(N-1) + 1$

Question NO 13

In the post-order traversal of a binary search tree, nodes process as:

Select Correct Option

- ✓ **Left-subtree, Right-subtree, Root.....(For More Visit VUStudents.pk)**
- ✓ Right-subtree, Root, Left-subtree
- ✓ Left-subtree, Root, Right-subtree
- ✓ Right-subtree, Left-subtree, Root

Question NO 14

The simplest case in a BST to delete a node is:

Select Correct Option

- ✓ When the node, that is to be deleted is root node
- ✓ When the node, that is to be deleted has both left and right child
- ✓ When the node, that is to be deleted has only one child
- ✓ **When the node, that is to be deleted is a leaf node...(For More Visit VUStudents.pk)**

Question NO 15

If class A defines class B as its friend, then:

Select Correct Option

- ✓ Class A can access private members of class B
- ✓ Class B can access only the public members of class A
- ✓ Class A can access only the public members of class B
- ✓ **Class B can access private members as class A.....(For More Visit VUStudents.pk)**





Question NO 16

In the statement `int& a=b;`

Select Correct Option

- ✓ a and b pointing to two different memory location
- ✓ a and b are two different names of the same memory location
- ✓ a and b are two different variables names
- ✓ **b hold the address of variable a**.....(For More Visit VUStudents.pk)

Question NO 17

The main use of AVL tree is:

Select Correct Option

- ✓ Searching of data
- ✓ **Storing of data**.....(For More Visit VUStudents.pk)
- ✓ Insertion of data
- ✓ Security of data

Question NO 18

In simple implementation of stack, `isFull()` method is used due to

Select Correct Option

- ✓ **Limitation of array**.....(For More Visit VUStudents.pk)
- ✓ Strength of array
- ✓ Linked list of connectivity
- ✓ Complexity of linked list

Question NO 19

Write in postfix form: $5 * (9 - 7)$

Select Correct Option

- ✓ $5 (9 7 -) *$
- ✓ $5 9 * 7 -$
- ✓ **$5 9 7 - *$**(For More Visit VUStudents.pk)
- ✓ $5 (9 7 -)$

Question NO 20

What will be result of following postfix expression?

$1 2 3 * + 2 -$

Select Correct Option

- ✓ 3
- ✓ **4**.....(For More Visit VUStudents.pk)
- ✓ 5
- ✓ 10





(File 2)

Question No 1

An array x of 100 integers is declared as.

Select Correct Option

- ✓ `int x = [100];`
- ✓ **`int x[100];`(For More Visit VUStudents.pk)**
- ✓ `integer array x = 100;`
- ✓ `integer x [] = 100;`

Question NO 2

What is the maximum depth of recursive calls a function may make?

Select Correct Option

- ✓ 1
- ✓ 2
- ✓ **n (where n is the argument)(For More Visit VUStudents.pk)**
- ✓ There is no fixed maximum

Question NO 3

A Linear Data Structure is the data structure in which data elements are arranged in a sequence or a linear list. Which of the following is Non Linear Data Structure?

Select Correct Option

- ✓ Arrays
- ✓ Linked Lists
- ✓ **Binary Search Trees.....(For More Visit VUStudents.pk)**
- ✓ Stack

Question NO 4

The new operation in C++ for dynamically allocating memory returns.

Select Correct Option

- ✓ Size of the memory it has allocated
- ✓ **Pointer to the memory it has allocated.....(For More Visit VUStudents.pk)**
- ✓ Both size and pointer to the memory it has allocated
- ✓ Value of the memory it has allocated

Question NO 5

In _____, a programmer uses two pointers in the node, i.e. one to point the next node and the other to point the previous node.

Select Correct Option

- ✓ **Linked list.....(For More Visit VUStudents.pk)**
- ✓ Doubly-linked list
- ✓ Array
- ✓ Structure





Question NO 6

The expression $AB+C^*$ is called?

Select Correct Option

- ✓ Prefix expression
- ✓ **Postfix expression**.....(For More Visit VUStudents.pk)
- ✓ Infix expression
- ✓ Prefix and infix expression

Question NO 7

Consider the function X as under
int X (int& Value)

```
{  
Return Value;  
}
```

Now a and b are integers in a calling function. Which one of the following is a valid call to the above function X?

Select Correct Option

- ✓ a = X (b);
- ✓ a = X (&b);
- ✓ **a = X (*b);**(For More Visit VUStudents.pk)
- ✓ a = X (&*b);

Question NO 8

Each Node in singly link list,

Select Correct Option

- ✓ **1 pointer**.....(For More Visit VUStudents.pk)
- ✓ 2 pointers
- ✓ 3 pointers
- ✓ 4 pointers

Question NO 9

Suppose n is the number of nodes in a complete Binary Tree, then maximum steps required for a search operation are

Select Correct Option

- ✓ **$\log_2 (n + 1) - 1$**(For More Visit VUStudents.pk)
- ✓ $\log_2 (n + 1)$
- ✓ $\log_2 (n) - 1$
- ✓ $\log_2 (n)$





Question NO 10

The next field in the last node of a singly-linked list is set to _____.

Select Correct Option

- ✓ NAN
- ✓ 1
- ✓ **NULL**.....(For More Visit VUStudents.pk)
- ✓ -1

Question NO 11

Consider the following function:

```
void test_a(int n)
{
    cout << n << " ";
    if (n>0)
```

Select Correct Option

- ✓ 4 2 0
- ✓ 0 2 4
- ✓ 4 2
- ✓ 2 4

Question NO 12

A queue where the dequeue operation does not depend upon FIFO, is called:

Select Correct Option

- ✓ Enqueue
- ✓ Simple queue
- ✓ Stack
- ✓ **Priority queue**.....(For More Visit VUStudents.pk)

Question NO 13

$a * (b+c)-d$ is an example of ----- expression.

Select Correct Option

- ✓ **infix**.....(For More Visit VUStudents.pk)
- ✓ prefix
- ✓ postfix
- ✓ allfix

Question NO 14

Below is a Binary Search Tree (BST). If we delete the value 50 from the root node, what would be the value in the root of the remaining tree?

Select Correct Option

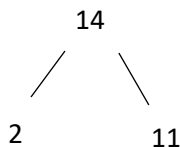
- ✓ 50
- ✓ 60
- ✓ 70
- ✓ 80





Question NO 15

Consider the following tree:



Select Correct Option

- ✓ 2
- ✓ 4
- ✓ 6
- ✓ 9

Question NO 16

Which one of the following calling methods does not change the original value of the argument in the calling function?

Select Correct Option

- ✓ **Call by passing the value of the argument**.....(For More Visit VUStudents.pk)
- ✓ Call by passing reference of the argument
- ✓ Call by passing the address of the argument
- ✓ Call by passing pointer of the argument

Question NO 17

Which of the following statement is NOT true for reference variable?

Select Correct Option

- ✓ Once a reference is created, it cannot be later made to reference another object.
- ✓ References cannot be NULL
- ✓ **References can be initialized**.....(For More Visit VUStudents.pk)
- ✓ It is not possible to refer directly to a reference object after it is defined.

Question NO 18

Searching an element in an AVL tree takes maximum _____ time (when n is number of nodes in AVL tree)

Select Correct Option

- ✓ $\log_2(n+1)$
- ✓ $\log_2(n+1)-1$
- ✓ **$1.44 \log_2 n$** (For More Visit VUStudents.pk)
- ✓ $1.66 \log_2 n$





Question NO 19

In the post-order traversal of a binary search tree, nodes process as:

Select Correct Option

- ✓ **Left-subtree, Right-subtree, Root**.....(For More Visit VUStudents.pk)
- ✓ Right-subtree, Root, Left-subtree
- ✓ Left-subtree, Root, Right-subtree
- ✓ Right-subtree, Left-subtree, Root

Question NO 20

The main use of AVL tree is:

Select Correct Option

- ✓ Searching of data
- ✓ **Storing of data**.....(For More Visit VUStudents.pk)
- ✓ Insertion of data
- ✓ Security of data

(File 3)

Question No 1

Each operator in a postfix expression refers to the previous _____ operand(s).

Select Correct Option

- ✓ **One**.....(For More Visit VUStudents.pk)
- ✓ Two
- ✓ Three
- ✓ Four

Question NO 2

Which one of the following statements is correct?

Select Correct Option

- ✓ **Array size is fixed once it is created**.....(For More Visit VUStudents.pk)
- ✓ Link List size is fixed once it is created
- ✓ Binary Search Tree size is fixed once it is created
- ✓ AVL Tree size is fixed once it is created

Question NO 3

Linked List always contains element that can be describe as,

Select Correct Option

- ✓ Redundant
- ✓ Recursive
- ✓ **Self-referential**.....(For More Visit VUStudents.pk)
- ✓ Bidirectional





Question NO 4

One difference between a queue and a stack is:

Select Correct Option

- ✓ Queues require dynamic memory, but stacks do not
- ✓ Stack require dynamic memory, but queues do not
- ✓ **Queues use two ends of the structure; stacks use only one**
- ✓ Stacks use two ends of the structure; queues use only one.

Question NO 5

Stack and Queue can be implemented using _____.

Select Correct Option

- ✓ **Singly Link List**.....(For More Visit VUStudents.pk)
- ✓ Binary Tree
- ✓ Binary Search Tree
- ✓ AVL Tree

Question NO 6

New items are added at the _____ of the stack.

Select Correct Option

- ✓ Bottom
- ✓ Middle
- ✓ **Top**.....(For More Visit VUStudents.pk)
- ✓ Center

Question NO 7

_____ is the stack characteristic but _____ was implemented because of the size limitation of the array.

Select Correct Option

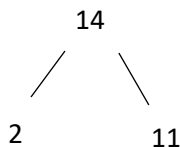
- ✓ isFull(), isEmpty()
- ✓ pop(), push()
- ✓ **isEmpty(), isFull()**.....(For More Visit VUStudents.pk)
- ✓ push(), pop()





Question NO 8

Consider the following tree:



Select Correct Option

- ✓ 2
- ✓ 4
- ✓ 6
- ✓ 9

Question NO 9

The difference between a “Binary Tree (BT)” and a “Binary Search Tree (BST)” is that,

Select Correct Option

- ✓ A BST has two children per node whereas a BT can have none, one, or two children per node
- ✓ **In BST, nodes are inserted based on the value they contain**
- ✓ In BT, nodes are inserted based on the value they contain
- ✓ There is no difference

Question NO 10

Which of the following operations returns “most recently entered value” from the stack?

Select Correct Option

- ✓ Push
- ✓ Recent
- ✓ **Top.....(For More Visit VUStudents.pk)**
- ✓ First

Question NO 11

Which of the following is correct about AVL Tree?

Select Correct Option

- ✓ It is identical to BCS except height of the left and right subtrees can differ by at least 1.
- ✓ It is identical to BCS except height of the left and right subtrees must be differ by at least 1
- ✓ It is not identical to BST, its totally different kind of tree.
- ✓ **It is identical to BCS except height of the left and right subtrees can be differ by at most 1**





Question NO 12

What will be postfix expression of the following infix expression?

Infix Expression: $a+b*c-d$

Select Correct Option

- ✓ $ab+c*d-$
- ✓ **$abc*+d-$**(For More Visit [VUStudents.pk](https://www.vustudents.pk))
- ✓ $abc+*d-$
- ✓ $abcd+*-$

Question NO 13

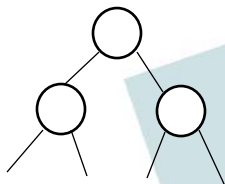
In the call by methodology, a copy of the object is passed to the called function.

Select Correct Option

- ✓ Reference
- ✓ **Value**.....(For More Visit [VUStudents.pk](https://www.vustudents.pk))
- ✓ Reference & Value
- ✓ Copy of the object can not be passed

Question NO 14

Consider the following tree, how many levels does it has?

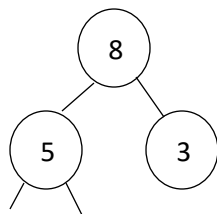


Select Correct Option

- ✓ One
- ✓ Two
- ✓ Three
- ✓ Seven

Question NO 15

Consider the following tree, which statement is correct about this tree.



Select Correct Option

- ✓ 3 and 5 are on the same level
- ✓ 7 is the parent of 3
- ✓ 9 is the root of this tree
- ✓ 5 is one the second level





For More Visit [VUStudents.pk](https://www.vustudents.pk)



Question NO 16

Recursive call of a function use data structure

Select Correct Option

- ✓ Linked List
- ✓ Queue
- ✓ **Stack**.....(For More Visit [VUStudents.pk](https://www.vustudents.pk))
- ✓ Table

Question NO 17

In the statement `int& a=b;`

Select Correct Option

- ✓ a and b pointing to two different memory location
- ✓ a and b are two different names of the same memory location
- ✓ a and b are two different variables names
- ✓ **b hold the address of variable a**.....(For More Visit [VUStudents.pk](https://www.vustudents.pk))

Question NO 18

The variable which are destroyed automatically when a function's execution ends are:

Select Correct Option

- ✓ Global variables
- ✓ **Local variables defined inside function body**.....(For More Visit [VUStudents.pk](https://www.vustudents.pk))
- ✓ Variable (objects) defined inside function body dynamically
- ✓ Variable (objects) defines inside function body statically

Question NO 19

The worst case of searching in binary search tree (BST) is:

Select Correct Option

- ✓ **When the data inserted in BST is sorted**.....(For More Visit [VUStudents.pk](https://www.vustudents.pk))
- ✓ When the height of left sub-tree is greater than right sub-tree
- ✓ When the height of right sub-tree is greater than left sub-tree
- ✓ When the tree is balanced

Question NO 20

In simple implementation of stack, `isFull()` method is used due to

Select Correct Option

- ✓ **Limitation of array**.....(For More Visit [VUStudents.pk](https://www.vustudents.pk))
- ✓ Strength of array
- ✓ Linked list of connectivity
- ✓ Complexity of linked list

The END

For More Visit [VUStudents.pk](https://www.vustudents.pk)



DATA STRUCTURE (CS301)

MID TERM MEGA FILE

❖ Which one of the following statement is NOT correct .

- ▶ In linked list the elements are necessarily to be contiguous
- ▶ **In linked list the elements may locate at far positions in the memory**
- ▶ In linked list each element also has the next to it
- ▶ In an array the elements are contiguous

❖ Each operator in a postfix expression refers to the previous _____ operand(s).

- ▶ One
- ▶ **Two**
- ▶ Three
- ▶ Four

❖ Which one of the following calling methods does not change the original value of the argument in the calling function?

- ▶ None of the given options
- ▶ **Call by passing the value of the argument**
- ▶ Call by passing reference of the argument
- ▶ Call by passing the address of the argument

❖ A tree is an AVL tree if

- ▶ Any one node fulfills the AVL condition
- ▶ At least half of the nodes fulfill the AVL condition
- ▶ All the nodes fulfill the AVL condition
- ▶ **None of the given options**

❖ Suppose currentNode refers to a node in a linked list (using the Node class with member variables called data and nextNode). What statement changes currentNode so that it refers to the next node?

- ▶ currentNode ++;
- ▶ currentNode = nextNode;
- ▶ currentNode += nextNode;
- ▶ **currentNode = currentNode->nextNode;**

❖ A queue where the de-queue operation depends not on FIFO, is called a priority queue

- ▶ False
- ▶ **True**

❖ Which one is a self- referential data type?

- ▶ Stack
- ▶ Queue
- ▶ **Link list**
- ▶ All of these

❖ Each node in doubly link list has,

- ▶ 1 pointer
- ▶ **2 pointers**
- ▶ 3 pointers
- ▶ 4 pointers

❖ I have implemented the queue with a linked list, keeping track of a front pointer and a rear pointer. Which of these pointers will change during an insertion into an *EMPTY* queue?

- ▶ Neither changes
- ▶ Only front pointer changes.
- ▶ **Only rear pointer changes.**
- ▶ Both change.

❖ Consider the following tree.



How many of the nodes have at least one sibling?

- ▶ 8
- ▶ 7
- ▶ **5**
- ▶ 6

❖ The nodes with no successor are called _____

- ▶ Root Nodes
- ▶ **Leaf Nodes**
- ▶ Both of these
- ▶ None of these

❖ AVL Tree is,

- ▶ **Non Linear data structure**
- ▶ Linear data structure
- ▶ Hybrid data structure (Mixture of Linear and Non Linear)
- ▶ None of the given options.

❖ We access elements in AVL Tree in,

- ▶ Linear way only
- ▶ **Non Linear way only**
- ▶ Both linear and non linear ways
- ▶ None of the given options.

❖ A binary search tree should have minimum of _____ node/s at each level,

- ▶ **One**
- ▶ Two
- ▶ Three
- ▶ Four

❖ Consider the following statements.

- (i) A binary tree can contain at least 2^L Nodes at level L.
- (ii) A complete binary tree of depth d is a binary tree that contains 2^L Nodes at each level L between 0 and d, both inclusive.
- (iii) The total number of nodes (T_n) in a complete binary tree of depth d is $2^{d+1} - 1$.
- (iv) The height of the complete binary tree can be written as $h = \log_2 (T_n + 1) - 1$ where T_n is Total number of Nodes.

Which one of the following is correct in respect of the above statements regarding the Binary trees?

- ▶ (i) and (iii) only
- ▶ (i), (ii) and (iii) only
- ▶ (ii) and (iii) only
- ▶ **(ii), (iii) and (iv) only**

❖ “+” is a _____ operator.

- ▶ Unary
- ▶ **Binary**
- ▶ Ternary
- ▶ None of the above

❖ A queue where the de-queue operation depends not on FIFO, is called a priority queue

- ▶ False
- ▶ **True**

❖ The data of the problem is of 2GB and the hard disk is of 1GB capacity, to solve this problem we should

- ▶ Use better data structures
- ▶ **Increase the hard disk space**
- ▶ Use the better algorithm
- ▶ Use as much data as we can store on the hard disk

❖ Consider the function X as under

```
int X (int& Value)
{
    return Value;
}
```


Now **a** and **b** are integers in a calling function. Which one of the following is a valid call to the above function **X**.

- ▶ `a = X (b) ;`
- ▶ `a = X (&b) ;`
- ▶ **`a = X (*b) ;`**
- ▶ None of the given options

❖ In the call by value methodology, a copy of the object is passed to the called function.

- ▶ False
- ▶ **True**

❖ The tree data structure is a

- ▶ Linear data structure
- ▶ **Non-linear data structure**
- ▶ Graphical data structure
- ▶ Data structure like queue

❖ When should you use a const reference parameter?

- ▶ Whenever the parameter has huge size
- ▶ Whenever the parameter has huge size, the function changes the parameter within its body, and you do NOT want these changes to alter the actual argument.
- ▶ Whenever the parameter has huge size, the function changes the parameter within its body, and you DO want these changes to alter the actual argument.
- ▶ **Whenever the parameter has huge size, and the function does not change the parameter within its body.**

❖ Here is the start of a C++ class declaration:

```
class foo
{
    public:
    void x(foo f);
    void y(const foo f);
    void z(foo f) const;

```

...

Which of the three member functions can alter the PRIVATE member variables of the foo object that activates the function?

- ▶ **Only x can alter the private member variables of the object that activates the function. (R)**
- ▶ Only y can alter the private member variables of the object that activates the function.
- ▶ Only z can alter the private member variables of the object that activates the function.
- ▶ Two of the functions can alter the private member variables of the object that activates the function.

❖ What is the maximum depth of recursive calls a function may make?

- ▶ 1
- ▶ 2
- ▶ **n (where n is the argument)**
- ▶ There is no fixed maximum

❖ Suppose n is the number of nodes in a complete Binary Tree then maximum steps required for a search operation are,

- ▶ **$\log_2 (n+1) - 1$**
- ▶ $\log_2 (n+1)$
- ▶ $\log_2 (n) - 1$
- ▶ $\log_2 (n)$

❖ In the linked list implementation of the stack class, where does the push member function places the new entry on the linked list?

- ▶ **At the head**
- ▶ At the tail
- ▶ After all other entries that are greater than the new entry.
- ▶ After all other entries that are smaller than the new entry.

❖ Suppose we have a *circular* array implementation of the queue class, with ten items in the queue stored at `data[2]` through `data[11]`. The CAPACITY is 42, i.e., the array has been declared to be of size 42. Where does the push member function place the new entry in the array?

- ▶ `data[1]`
- ▶ `data[2]`
- ▶ **`data[11]`**
- ▶ `data[12]`

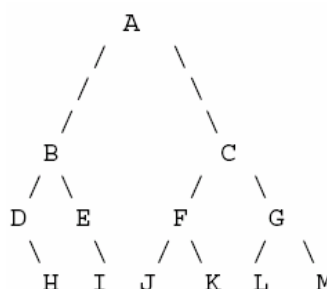
❖ The expression $AB+C^*$ is called ?

- ▶ Prefix expression
- ▶ **Postfix expression**
- ▶ Infix expression
- ▶ None of these

❖ _____ is a binary tree where every node has a value, every node's left subtree contains only values less than or equal to the node's value, and every node's right subtree contains only values that are greater than or equal ?

- ▶ **Strictly Binary Tree**
- ▶ Binary Search tree
- ▶ AVL tree
- ▶ All of these

❖ Consider the following binary search tree (BST):



If node A in the BST is deleted, which two nodes are the candidates to take its place?

- ▶ **J and I (Pending)**

- ▶ H and E
- ▶ D and E
- ▶ L and M

❖ **Let's call the node as a that requires re-balancing. Consider the two cases given below:**

- 1) An insertion into left subtree of the left child of a
- 2) An insertion into right subtree of the right child of a.

Which of the following statement is correct about these two cases.

- ▶ The insertion occurs outside (i.e., left-left or right-right) in cases 1 and 2. single rotation can fix the balance in these two cases.
- ▶ The insertion occurs inside ((i.e., left-left or right-right) in cases 1 and 2. single rotation cannot fix the balance in these two cases

❖ **We access elements in AVL Tree in,**

- ▶ Linear way only
- ▶ **Non Linear way only**
- ▶ Both linear and non linear ways
- ▶ None of the given options.

❖ **AVL Tree is,**

- ▶ **Non Linear data structure**
- ▶ Linear data structure
- ▶ Hybrid data structure (Mixture of Linear and Non Linear)
- ▶ None of the given options.

❖ **Which one of the following is a valid postfix expression?**

- ▶ **ab+c*d-**
- ▶ abc*+d-
- ▶ abc+*d-
- ▶ (abc*)+d-

❖ **The tree data structure is a**

- ▶ Linear data structure
- ▶ **Non-linear data structure**
- ▶ Graphical data structure
- ▶ Data structure like queue

❖ **A Compound Data Structure is the data structure which can have multiple data items of same type or of different types. Which of the following can be considered compound data structure?**

- ▶ Arrays
- ▶ LinkLists
- ▶ **Binary Search Trees**
- ▶ All of the given options

❖ **Suppose a pointer has been declared in main but has not assigned any variable address then**

- ▶ ☐ That pointer points to First byte in main function
- ▶ **That pointer contains a NULL value**
- ▶ ☐ None of these
- ▶ That pointer points to any memory address

❖ Here is the start of a C++ class declaration:

```
class foo
{
    public:
    void x(foo f);
    void y(const foo f);
    void z(foo f) const;
```

Which of the three member functions can alter the PRIVATE member variables of the foo object that activates the function?

▶ **Only x can alter the private member variables of the object that activates the function.**

▶ Only y can alter the private member variables of the object that activates the function.

▶ ☐ Only z can alter the private member variables of the object that activates the function.

▶ Two of the functions can alter the private member variables of the object that activates the function.

❖ **The operation for removing an entry from a stack is traditionally called:**

▶ delete

▶ peek

▶ **pop**

▶ remove

❖ **Which statement of the following statements is incorrect?**

▶ Lists can be implemented by using arrays or linked lists

▶ A list is a sequence of one or more data items

▶ **Stack is a special kind of list in which all insertions and deletions take place at one end**

▶ Stacks are easier to implement than lists

❖ **Parameters in function call are passed using,**

▶ Stack

▶ **Queue**

▶ Binary Search Tree

▶ AVL Tree

❖ **Consider the following sequence of push operations in a stack:**

```
stack.push('7');
stack.push('8');
stack.push('9');
stack.push('10');
stack.push('11');
stack.push('12');
```

▶ ☐ **7 8 9 10 11 12**

▶ 9 8 11 10 7 12

▶ 9 10 8 11 12 7

▶ 9 10 8 12 7 11

❖ What is the maximum depth of recursive calls a function may make?

- ▶ 1
- ▶ 2

▶ **n (where n is the argument)**

- ▶ There is no fixed maximum

❖ Consider the following function:

```
void test_a(int n)
{
    cout << n << " ";
    if (n>0)
        test_a(n-2);
}
```

What is printed by the call test_a(4)?

- ▶ 4 2
- ▶ 0 2 4
- ▶ 0 2
- ▶ **2 4**

❖ Queue follows,

- ▶ Last in First out
- ▶ First in Last out
- ▶ **First in First out**
- ▶ None of these

❖ _____ is a binary tree where every node has a value, every node's left subtree contains only values less than or equal to the node's value, and every node's right subtree contains only values that are greater than or equal ?

▶ **Strictly Binary Tree**

- ▶ Binary Search tree
- ▶ AVL tree
- ▶ All of these

❖ Four statements about trees are below. Three of them are correct. Which one is INCORRECT?

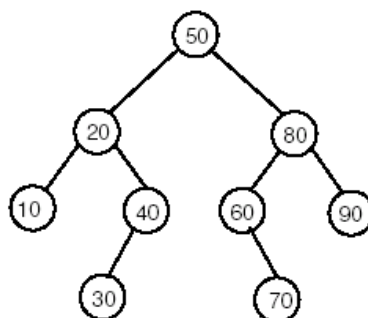
▶ ☐ **Trees are recursively defined multi-dimensional data structures**

▶ The order of a tree indicates a maximum number of children allowed at each node of the tree

▶ A search tree is a special type of tree where all values (i.e. keys) are ordered

▶ If Tree1's size is greater than Tree2's size, then the height of Tree1 must also be greater than Tree2's height.

❖ Below is a binary search tree. If we delete the value 50 using the algorithm we discussed, what value will be in the root of the remaining tree?



- ▶ 50
- ▶ 60
- ▶ 70
- ▶ **80**

❖ _____ is a data structure that can grow easily dynamically at run time without having to copy existing elements.

- ▶ Array
- ▶ **List**
- ▶ Both of these
- ▶ None of these

❖ _____ **only removes items in reverse order as they were entered.**

- ▶ **Stack**
- ▶ Queue
- ▶ Both of these
- ▶ None of these

❖ **Here is a small function definition:**

```
void f(int i, int &k)
{
    i = 1;
    k = 2;
}
```

Suppose that a main program has two integer variables x and y, which are given the value 0. Then the main program calls f(x,y); What are the values of x and y after the function f finishes?

- ▶ Both x and y are still 0.
- ▶ x is now 1, but y is still 0.
- ▶ **x is still 0, but y is now 2.**
- ▶ x is now 1, and y is now 2.

❖ **Select the one *FALSE* statement about binary trees:**

- ▶ **Every binary tree has at least one node.**
- ▶ Every non-empty tree has exactly one root node.
- ▶ Every node has at most two children.
- ▶ Every non-root node has exactly one parent.

❖ **Every AVL is _____**

- ▶ Binary Tree
- ▶ Complete Binary Tree
- ▶ None of these
- ▶ **Binary Search Tree**

❖ **Searching an element in an AVL tree take maximum _____ time (where n is no. of nodes in AVL tree),**

- ▶ $\log_2(n+1)$
- ▶ $\log_2(n+1) - 1$

► **1.44 $\log_2 n$**

► $1.66 \log_2 n$

❖ Suppose that we have implemented a *priority queue* by storing the items in a heap. We are now executing a reheapification downward and the out-of-place node has priority of 42. The node's parent has a priority of 72, the left child has priority 52 and the node's right child has priority 62. Which statement best describes the status of the reheapification.

- The reheapification is done.
- The next step will interchange the two children of the out-of-place node.
- The next step will swap the out-of-place node with its parent.
- The next step will swap the out-of-place node with its left child.

❖ Suppose you implement a heap (with the largest element on top) in an array. Consider the different arrays below, determine the one that *cannot* possibly be a heap:

- 7 6 5 4 3 2 1
- 7 3 6 2 1 4 5
- 7 6 4 3 5 2 1
- 7 3 6 4 2 5 1

❖ If there are 23 external nodes in a binary tree then what will be the no. of internal nodes in this binary tree?

- 23
- 24
- 21
- **22**

❖ If there are N external nodes in a binary tree then what will be the no. of internal nodes in this binary tree?

- **$N - 1$**
- $N + 1$
- $N + 2$
- N

❖ Which one of the following is NOT the property of equivalence relation:

- Reflexive
- Symmetric
- Transitive
- **Associative**

❖ The definition of Transitivity property is

- For all element x member of S , $x R x$
- For all elements x and y , $x R y$ if and only if $y R x$
- **For all elements x , y and z , if $x R y$ and $y R z$ then $x R z$**
- For all elements w , x , y and z , if $x R y$ and $w R z$ then $x R z$

❖ Union is a _____ time operation.

- **Constant**
- Polynomial
- Exponential
- None of the given options

❖ Which of the following is NOT a correct statement about Table ADT.

- ▶ In a table, the type of information in columns may be different.
- ▶ A table consists of several columns, known as entities.
- ▶ The row of a table is called a record.
- ▶ A major use of table is in databases where we build and use tables for keeping information.

❖ In the worst case of deletion in AVL tree requires _____.

- ▶ Only one rotation
- ▶ Rotation at each non-leaf node
- ▶ Rotation at each leaf node
- ▶ Rotations equal to $\log_2 N$

❖ Binary Search is an algorithm of searching, used with the _____ data.

- ▶ Sorted
- ▶ Unsorted
- ▶ Heterogeneous
- ▶ Random

❖ Which of the following statement is correct?

- ▶ A Threaded Binary Tree is a binary tree in which every node that does not have a left child has a THREAD (in actual sense, a link) to its INORDER successor.
- ▶ A Threaded Binary Tree is a binary tree in which every node that does not have a right child has a THREAD (in actual sense, a link) to its PREORDER successor.
- ▶ A Threaded Binary Tree is a binary tree in which every node that does not have a right child has a THREAD (in actual sense, a link) to its INORDER successor.
- ▶ A Threaded Binary Tree is a binary tree in which every node that does not have a right child has a THREAD (in actual sense, a link) to its POSTORDER successor.

❖ By using _____ we avoid the recursive method of traversing a Tree, which makes use of stacks and consumes a lot of memory and time.

- ▶ Binary tree only
- ▶ Threaded binary tree
- ▶ Heap data structure
- ▶ Huffman encoding

❖ Which of the following statement is NOT true about threaded binary tree?

- ▶ Right thread of the right-most node points to the *dummy* node.
- ▶ Left thread of the left-most node points to the *dummy* node.
- ▶ The left pointer of dummy node points to the root node of the tree.
- ▶ Left thread of the right-most node points to the *dummy* node.

❖ Consider a min heap, represented by the following array:

11,22,33,44,55 After inserting a node with value 66. Which of the following is the updated min heap?

- ▶ 11,22,33,44,55,66
- ▶ 11,22,33,44,66,55
- ▶ 11,22,33,66,44,55
- ▶ 11,22,66,33,44,55

- ❖ Consider a min heap, represented by the following array:
3,4,6,7,5 After calling the function deleteMin(). Which of the following is the updated min heap?

- ▶ 4,6,7,5
- ▶ 6,7,5,4
- ▶ 4,5,6,7
- ▶ 4,6,5,7

- ❖ We can build a heap in _____ time.

▶ **Linear**

- ▶ Exponential
- ▶ Polynomial
- ▶ None of the given options

- ❖ Suppose we are sorting an array of eight integers using quick sort, and we have just finished the first partitioning with the array looking like this:

2 5 1 7 9 12 11 10

Which statement is correct?

▶ **The pivot could be either the 7 or the 9.**

- ▶ The pivot could be the 7, but it is not the 9.
- ▶ The pivot is not the 7, but it could be the 9.
- ▶ Neither the 7 nor the 9 is the pivot.

- ❖ Which formula is the best approximation for the depth of a heap with n nodes?

▶ **$\log$ (base 2) of n**

- ▶ The number of digits in n (base 10), e.g., 145 has three digits
- ▶ The square root of n
- ▶ n

- ❖ Suppose you implement a Min heap (with the smallest element on top) in an array. Consider the different arrays below; determine the one that *cannot* possibly be a heap:

- ▶ 16, 18, 20, 22, 24, 28, 30
- ▶ 16, 20, 18, 24, 22, 30, 28
- ▶ 16, 24, 18, 28, 30, 20, 22
- ▶ 16, 24, 20, 30, 28, 18, 22

- ❖ While joining nodes in the building of Huffman encoding tree if there are more nodes with same frequency, we choose the nodes _____.

▶ **Randomly**

- ▶ That occur first in the text message
- ▶ That are lexically smaller among others.
- ▶ That are lexically greater among others

- ❖ Consider the following paragraph with blanks.

A is a linear list where and take place at the same end . This end is called the

What would be the correct filling the above blank positions?

- ▶ (i) queue (ii) insertion (iii) removals (iv) top

- ▶ (i) stack (ii) insertion (iii) removals (iv) bottom
- ▶ **(i) stack (ii) insertion (iii) removals (iv) top**
- ▶ (i) tree (ii) insertion (iii) removals (iv) top

❖ A binary tree with 33 internal nodes has _____ links to internal nodes.

- ▶ 31
- ▶ **32**
- ▶ 33
- ▶ 66

❖ Which traversal gives a decreasing order of elements in a heap where the max element is stored at the top?

- ▶ post-order
- ▶ level-order
- ▶ inorder
- ▶ None of the given options

❖ What requirement is placed on an array, so that *binary search* may be used to locate an entry?

- ▶ The array elements must form a heap.
- ▶ The array must have at least 2 entries.
- ▶ The array must be sorted.
- ▶ The array's size must be a power of two.

❖ Which of the following is a non linear data structure?

- ▶ Linked List
- ▶ Stack
- ▶ Queue
- ▶ **Tree**

Question No: 1 (Marks: 1) - Please choose one

In the statement `int x[6];`, we cannot assign any value to x because x is not an lvalue.

- ▶ True
- ▶ **False**

Question No: 2 (Marks: 1) - Please choose one

What will be postfix expression of the following infix expression?

Infix Expression : $a+b*c-d$

- ▶ $ab+c*d-$
- ▶ **$abc*+d-$**
- ▶ $abc+*d-$
- ▶ $abcd+*-$

Question No: 3 (Marks: 1) - Please choose one

Which one of the following calling methods does not change the original value of the argument in the calling function?

- ▶ None of the given options

- ▶ Call by passing the value of the argument
- ▶ Call by passing reference of the argument
- ▶ **Call by passing the address of the argument**

Question No: 4 (Marks: 1) - Please choose one
In a program a reference variable, say x, can be declared as

- ▶ **int &x ;**
- ▶ int *x ;
- ▶ int x ;
- ▶ None of the given options

Question No: 5 (Marks: 1) - Please choose one

A tree is an AVL tree if

- ▶ None of the given options

Question No: 6 (Marks: 1) - Please choose one

Consider the following pseudo code
 declare a stack of characters

```
while ( there are more characters in the word to read )
{
  read a character
  push the character on the stack
}
while ( the stack is not empty )
{
  pop a character off the stack
  write the character to the screen
}
```

What is written to the screen for the input "apples"?

- ▶ selpa
- ▶ **selppa**
- ▶ apples
- ▶ aaappppplleess

Question No: 7 (Marks: 1) - Please choose one

In the following C++ code, how many function calls are made?

```
int x, y, z;
x = 2;
y = 3 + x;
z = foobar(x,y);
```

- ▶ 1
- ▶ 4
- ▶ 7
- ▶ **8**

Question No: 8 (Marks: 1) - Please choose one

We can add elements in QUEUE From _____

- ▶ **Front**
- ▶ Rear
- ▶ From Both Rare and Front

- ▶ None of these

Question No: 9 (Marks: 1) - Please choose one
Consider the following tree.

How many of the nodes have at least one sibling?

- ▶ 8
- ▶ 7
- ▶ 5
- ▶ 6

Question No: 10 (Marks: 1) - Please choose one
Consider the following tree.

How many descendants does the root have?

- ▶ 5
- ▶ 6
- ▶ 7
- ▶ 8

Question No: 11 (Marks: 1) - Please choose one
Below is a binary search tree. If we delete the value 50 using the algorithm we discussed, what value will be in the root of the remaining tree?

- ▶ 50
- ▶ 60
- ▶ 70
- ▶ 80

Question No: 12 (Marks: 1) - Please choose one
We access elements in AVL Tree in,

- ▶ Linear way only
- ▶ **Non Linear way only**
- ▶ Both linear and non linear ways
- ▶ None of the given options.

Question No: 13 (Marks: 1) - Please choose one

Which of the following statement regarding binary tree is NOT correct.

- ▶ A binary tree can contain at least 2^L Nodes at level L.
- ▶ A complete binary tree of depth d is a binary tree that contains 2^L Nodes at each level L between 0 and d, both inclusive.
- ▶ The total number of nodes (T_n) in a complete binary tree of depth d is $2^{d+1} - 1$.
- ▶ **The height of the complete binary tree can be written as $h = \log_2 (T_n + 1) - 1$ where T_n is Total number of Nodes.**

Question No: 14 (Marks: 1) - Please choose one
The following are statements related to queues.

- i. The last item to be added to a queue is the first item to be removed
- ii. A queue is a structure in which both ends are not used
- iii. The last element hasn't to wait until all elements preceding it on the queue are removed
- iv. A queue is said to be a last-in-first-out list or LIFO data structure.

Which of the above is/are related to normal queues?

- ▶ **(iii) and (ii) only**
- ▶ (i), (ii) and (iv) only

- ▶ (ii) and (iv) only
- ▶ None of the given options

Question No: 15 (Marks: 1) - Please choose one

The _____ method of list data structure removes the element residing at the current position.

- ▶ Add
- ▶ next
- ▶ **remove**
- ▶ find

Question No: 16 (Marks: 1) - Please choose one

it will be efficient to place stack elements at the start of the list because insertion and removal take _____ time.

- ▶ Variable
- ▶ **Constant**
- ▶ Inconsistent
- ▶ None of the above

Question No: 1 (Marks: 1) - Please choose one

Which one of the following statement is NOT correct .

- ▶ In linked list the elements are necessarily to be contiguous
- ▶ **In linked list the elements may locate at far positions in the memory**
- ▶ In linked list each element also has the address of the element next to it
- ▶ In an array the elements are contiguous

Question No: 2 (Marks: 1) - Please choose one

In a program a reference variable, say x, can be declared as

- ▶ **int &x ;**
- ▶ int *x ;
- ▶ int x ;
- ▶ None of the given options

Question No: 3 (Marks: 1) - Please choose one

Linked lists are collections of data items "lined up in a row" , insertions and deletions can be made only at the front and the back of a linked list.

- ▶ True
- ▶ **False**

Question No: 4 (Marks: 1) - Please choose one

A Linear Data Structure is the data structure in which data elements are arranged in a sequence or a linear list. Which of the following is Non Linear Data Structure?

- ▶ Arrays
- ▶ LinkLists
- ▶ **Binary Search Trees**
- ▶ None of these

Question No: 5 (Marks: 1) - Please choose one

A queue where the de-queue operation depends not on FIFO, is called a priority queue

- ▶ False
- ▶ **True**

Question No: 6 (Marks: 1) - Please choose one

Which one of the following statements is correct?

- ▶ ☐ Array size is fixed once it is created.

- ▶ Link List size is fixed ☐ once it is created.
- ▶ Binary Search Tree size is ☐ ☐ fixed once it is created
- ▶ AVL Tree size is fixed ☐ once it is created

Question No: 7 (Marks: 1) - Please choose one

Which one of the following is correct about pointers?

- ▶ They always point to ☐ different memory locations
- ▶ They may point to a single ☐ ☐ memory location
- ☐ ▶ The address of two pointer variables is same
- ▶ None of these ☐

Question No: 8 (Marks: 1) - Please choose one

Which of the following abstract data types are NOT used by Integer Abstract Data type group?

- ▶ ☐ short
- ▶ int ☐
- ▶ float ☐
- ▶ long ☐ ☐

Question No: 9 (Marks: 1) - Please choose one

The operation for adding an entry to a stack is traditionally called :

- ▶ add ☐
- ▶ append ☐
- ▶ insert ☐ ☐
- ☐ ▶ push

Question No: 10 (Marks: 1) - Please choose one

The operation for removing an entry from a stack is traditionally called:

- ▶ delete ☐
- ▶ peek ☐
- ▶ ☐ pop
- ▶ remove ☐

Question No: 11 (Marks: 1) - Please choose one

We can add elements in QUEUE From _____

- ▶ Front ☐
- ▶ ☐ Rear
- ▶ From Both Rare and Front ☐
- ▶ None of these ☐
- ☐

Question No: 12 (Marks: 1) - Please choose one

The difference between a binary tree and a binary search tree is that ,

- ▶ a binary search tree has ☐ ☐ two children per node whereas a binary tree can have none, one, or two children per node
- ▶ ☐ in binary search tree nodes are inserted based on the values they contain
- ▶ in binary tree nodes are inserted based on the values they contain
- ▶ none of these ☐ ☐

Question No: 13 (Marks: 1) - Please choose one

Suppose n is the number of nodes in a complete Binary Tree then maximum steps required for a search operation are,

- ▶ ☐ $\log_2 (n+1) - 1$
- ▶ $\log_2 (n+1)$
- ▶ $\log_2 (n) - 1$
- ▶ $\log_2 (n)$

Question No: 14 (Marks: 1) - Please choose one

The following is a segment of a C program.

```
int pqr(BinaryNode t)
{ if (t == null )
return -1;
```

else

return 1+max(pqr(t.left),pqr(t.right)) }

Identify, what the above program intend(s) to do?

- ▶ Compute the height of a ☐ ☐ binary tree using an in-order traversal
- ▶ Compute the height of a ☐ ☐ binary tree using a pre-order traversal
- ▶ ☐ Compute the depth of a binary tree using a pre-order traversal
- ▶ Compute the depth of a ☐ ☐ binary tree using a postorder traversal

Question No: 15 (Marks: 1) - Please choose one

Consider the following infix expression:

$3 + 5 * 6 - 7 * (8 + 5)$

Which of the following is a correct equivalent expression(s) for the above?

- ▶ $3 \ 6 \ 5 \ + \ * \ 7 \ 5 \ 8 \ + \ - \ *$
- ▶ $3 \ 6 \ 5 \ 7 \ 5 \ 8 \ + \ * \ + \ - \ *$
- ▶ $3 \ 5 \ 6 \ + \ * \ 7 \ 8 \ 5 \ + \ - \ *$
- ▶ $3 \ 5 \ 6 \ * \ + \ 7 \ 8 \ 5 \ + \ * \ -$

Question No: 16 (Marks: 1) - Please choose one

An array is a group of consecutive related memory locations.

- ▶ True
- ▶ False

Question No: 17 (Marks: 1)

Is this a correct statement?

Give answer in Yes or No.

- ▶ A node cannot be deleted, when the node to be deleted has both left and right subtrees.
- ▶ No, it can be deleted.

Question No: 18 (Marks: 1)

Deleting a leaf node in binary search tree involves setting _____ pointer/s of that nodes parent as null.

- ▶ 1
- ▶ 2
- ▶ 3
- ▶ 4

Question No: 1 (Marks: 1) - Please choose one

A subscript of an array may be an integer or an integer expression.

- ▶ True
- ▶ False

Question No: 2 (Marks: 1) - Please choose one

Doubly Linked List always has one NULL pointer.

- ▶ True
- ▶ False

Question No: 3 (Marks: 1) - Please choose one

In which of the traversal method, the recursive calls can be used to traverse a binary tree ?

- ▶ In preorder traversal only
- ▶ In inorder traversal only
- ▶ In postorder traversal only
- ▶ All of the given options

Question No: 4 (Marks: 1) - Please choose one

A tree is an AVL tree if

- ▶ Any one node fulfills the AVL condition
- ▶ At least half of the nodes fulfill the AVL condition
- ▶ All the nodes fulfill the AVL condition
- ▶ None of the given options

Question No: 5 (Marks: 1) - Please choose one

Suppose currentNode refers to a node in a linked list (using the Node class with member variables called data and nextNode). What boolean expression will be true when cursor refers to the tail node of the list?

- ▶ (currentNode == null)
- ▶ (currentNode->nextNode == null)
- ▶ (nextNode.data == null)
- ▶ (currentNode.data == 0.0)

Question No: 6 (Marks: 1) - Please choose one

Suppose that the class declaration of SomeClass includes the following function prototype.
bool LessThan(SomeClass anotherObject);

Which of the following tests in the client code correctly compares two class objects alpha and beta?

- ▶ if (alpha < beta)
- ▶ if (alpha.LessThan(beta))
- ▶ if (LessThan(alpha, beta))
- ▶ if (LessThan(alpha).beta)

Question No: 7 (Marks: 1) - Please choose one

In C what is the operation that you can not do with primitive types?

- ▶ Assign a value to primitive type using a literal
- ▶ Declare primitive types to be constant using the Const keyword
- ▶ Create a new instance of primitive type with New keyword
- ▶ None of these

Question No: 8 (Marks: 1) - Please choose one

The operation for adding an entry to a stack is traditionally called :

- ▶ add
- ▶ append
- ▶ insert
- ▶ push

Question No: 9 (Marks: 1) - Please choose one

The operation for removing an entry from a stack is traditionally called:

- ▶ delete
- ▶ peek
- ▶ pop
- ▶ remove

Question No: 10 (Marks: 1) - Please choose one

Consider the following sequence of push operations in a stack:

```
stack.push('7');
stack.push('8');
stack.push('9');
stack.push('10');
stack.push('11');
stack.push('12');
```

- ▶ 7 8 9 10 11 12
- ▶ 9 8 11 10 7 12
- ▶ 9 10 8 11 12 7
- ▶ 9 10 8 12 7 11

Question No: 11 (Marks: 1) - Please choose one

_____ is the maximum number of nodes that you can have on a stack-linked list ?

- ▶ Zero

- ▶ $2n$ (where n is the number of nodes in linked list)
- ▶ Any Number
- ▶ None of these

Question No: 12 (Marks: 1) - Please choose one

Which of the following can be used to reverse a string value,

- ▶ Stack
- ▶ Queue
- ▶ Both of these
- ▶ None of these

Question No: 13 (Marks: 1) - Please choose one

Consider the following tree,

How many leaves does it have?

- ▶ 2
- ▶ 4
- ▶ 6
- ▶ 9

Question No: 14 (Marks: 1) - Please choose one

AVL Tree is,

- ▶ **Non Linear data structure**
- ▶ Linear data structure
- ▶ Hybrid data structure (Mixture of Linear and Non Linear)
- ▶ None of the given options.

Question No: 15 (Marks: 1) - Please choose one

The following are statements related to queues.

- (i) The last item to be added to a queue is the first item to be removed
- (ii) A queue is a structure in which both ends are not used
- (iii) The last element hasn't to wait until all elements preceding it on the queue are removed
- (iv) A queue is said to be a last-in-first-out list or LIFO data structure.

Which of the above is/are related to normal queues?

- ▶ (iii) and (ii) only
- ▶ (i), (ii) and (iv) only
- ▶ (ii) and (iv) only
- ▶ None of the given options

Question No: 16 (Marks: 1) - Please choose one

An array is a group of consecutive related memory locations.

- ▶ True
- ▶ False

Question No: 1 (Marks: 1) - Please choose one

In an array we can store data elements of different types.

- ▶ True
- ▶ **False**

Question No: 2 (Marks: 1) - Please choose one

In an array list the current element is

- ▶ The first element
- ▶ The middle element
- ▶ The last element
- ▶ **The element where the current pointer points to**

Question No: 3 (Marks: 1) - Please choose one

Which one of the following calling methods does not change the original value of the argument in the calling function?

- ▶ None of the given options
- ▶ **Call by passing the value of the argument**
- ▶ Call by passing reference of the argument
- ▶ Call by passing the address of the argument

Question No: 4 (Marks: 1) - Please choose one

Which one of the following statements is NOT correct?

- ▶ Array size can be changed after its creation.
- ▶ Link List size can be changed after its creation
- ▶ Binary Search Tree size can be changed after its creation
- ▶ AVL Tree size can be changed after its creation

Question No: 5 (Marks: 1) - Please choose one

Suppose that the class declaration of SomeClass includes the following function prototype.

`bool LessThan( SomeClass anotherObject );`

Which of the following tests in the client code correctly compares two class objects alpha and beta?

- ▶ `if (alpha < beta)`
- ▶ `if (alpha.LessThan(beta))`
- ▶ `if (LessThan(alpha, beta))`
- ▶ `if (LessThan(alpha).beta)`

Question No: 6 (Marks: 1) - Please choose one

A queue is a _____ data structure, whereas a stack is a _____ data structure.

- ▶ **FIFO, LIFO**
- ▶ LIFO, FIFO
- ▶ none of these
- ▶ both of these

Question No: 7 (Marks: 1) - Please choose one

Which one of the following operators has higher priority than all of others ?

- ▶ Multiplication operator
- ▶ Minus operator
- ▶ Plus operator
- ▶ Exponentiation operator

Question No: 8 (Marks: 1) - Please choose one

Each node in Binary Search Tree has

- ▶ 1 pointer
- ▶ 2 pointers

- ▶ 3 pointers
- ▶ 4 pointers

Question No: 9 (Marks: 1) - Please choose one

Four statements about trees are below. Three of them are correct. Which one is INCORRECT?

- ▶ Trees are recursively defined multi-dimensional data structures
- ▶ The order of a tree indicates a maximum number of children allowed at each node of the tree
- ▶ A search tree is a special type of tree where all values (i.e. keys) are ordered
- ▶ If Tree1's size is greater than Tree2's size, then the height of Tree1 must also be greater than Tree2's height.

Question No: 10 (Marks: 1) - Please choose one

Which of the following is "TRUE" about arrays,

- ▶ We can increase the size of arrays after their creation.
- ▶ We can decrease the size of arrays after their creation.
- ▶ We can increase but can't decrease the size of arrays after their creation.
- ▶ We can neither increase nor decrease the array size after their creation.

Question No: 11 (Marks: 1) - Please choose one

Searching an element in an AVL tree take maximum _____ time (where n is no. of nodes in AVL tree),

- ▶ $\log_2(n+1)$
- ▶ $\log_2(n+1) - 1$
- ▶ $1.44 \log_2 n$
- ▶ $1.66 \log_2 n$

Question No: 12 (Marks: 1) - Please choose one

There is/are _____ case/s for rotation in an AVL tree,

- ▶ 1
- ▶ 3
- ▶ 2
- ▶ 4

Question No: 13 (Marks: 1) - Please choose one

Consider the following statements.

- i. A binary tree can contain at least 2^L Nodes at level L.
- ii. A complete binary tree of depth d is a binary tree that contains 2^L Nodes at each level L between 0 and d, both inclusive.
- iii. The total number of nodes (T_n) in a complete binary tree of depth d is $2^{d+1} - 1$.
- iv. The height of the complete binary tree can be written as $h = \log_2 (T_n + 1) - 1$ where T_n is Total number of Nodes.

Which one of the following is correct in respect of the above statements regarding the Binary trees?

- ▶ (i) and (iii) only
- ▶ (i), (ii) and (iii) only
- ▶ (ii) and (iii) only
- ▶ (ii), (iii) and (iv) only

Question No: 14 (Marks: 1) - Please choose one

Consider the following infix expression.

$5 + 6/2$

If one converts the above expression into postfix, what would be the resultant expression?

- ▶ $56 / + 2$
- ▶ $5\ 6\ 2\ /\ +$
- ▶ $5\ 6\ /\ 2\ +$
- ▶ $/62\ +\ 5$

Question No: 15 (Marks: 1) - Please choose one

Which of the following is a non linear data structure?

- ▶ Linked List
- ▶ Stack
- ▶ Queue
- ▶ **Tree**

Question No: 16 (Marks: 1) - Please choose one

“+” is a _____ operator.

- ▶ Unary
- ▶ **Binary**
- ▶ Ternary
- ▶ None of the above

Question: (Marks: 1) - Please choose one

In a complete binary tree of depth 5 the number of non-leaf nodes is

- 15
- 32
- 16
- 31

Question: (Marks: 1) - Please choose one

Which of the following is NOT a linear data structure?

- Linked List
- Stack
- Queue
- **Tree**

Question: (Marks: 1) - Please choose one

Recursive function calls are implemented internally using a data structure

- Stack
- Link-List
- **Tree**
- Queue

Question: (Marks: 1) - Please choose one

We access elements in AVL Tree in,

- Linear way only
- Non Linear way only
- Both linear and non linear ways
- None of the given options.

Question: (Marks: 1) - Please choose one

Consider the following tree,

How many leaves does it have?

- 2
- 4
- 6
- 9

Question: (Marks: 1) - Please choose one

In the statement `int x[6];`, we cannot assign any value to x because x is not an lvalue.

- True
- **False**

Question: (Marks: 1) - Please choose one

In the following C++ code, how many function calls are made?

```
int x, y, z;
x = 2;
y = 3 + x;
z = foobar(x,y);
```

- 1
- 4
- 7
- 8

Question: (Marks: 1) - Please choose one

Consider the following infix expression:

$3 + 5 * 6 - 7 * (8 + 5)$

Which of the following is a correct equivalent expression(s) for the above?

- $6\ 5\ +\ *\ 7\ 5\ 8\ +\ -\ *$
- $6\ 5\ 7\ 5\ 8\ +\ *\ +\ -\ *$
- $5\ 6\ +\ *\ 7\ 8\ 5\ +\ -\ *$
- $3\ 5\ 6\ *\ +\ 7\ 8\ 5\ +\ *\ -$

Question: (Marks: 1) - Please choose one

A subscript of an array may be an integer or an integer expression.

- True
- False
-

Question: (Marks: 1) - Please choose one

Which of the following is "TRUE" about arrays,

- We can increase the size of arrays after their creation.
- We can decrease the size of arrays after their creation.
- We can increase but can't decrease the size of arrays after their creation.
- We can neither increase nor decrease the array size after their creation.

Question: (Marks: 1) - Please choose one

Searching an element in an AVL tree take maximum _____ time (where n is no. of nodes in AVL tree),

- $\log_2(n+1)$
- $\log_2(n+1) - 1$
- **$1.44 \log_2 n$**
- $1.66 \log_2 n$

Question: (Marks: 1) - Please choose one

There is/are _____ case/s for rotation in an AVL tree,

- 1
- 3
- 2
- 4

Question: (Marks: 1) - Please choose one

Consider the following infix expression.

$5 + 6/2$

If one converts the above expression into postfix, what would be the resultant expression?

- $56/ + 2$
- $5\ 6\ 2\ /\ +$
- $5\ 6\ /\ 2\ +$
- $/62 + 5$

Question No: 16 (Marks: 1) - Please choose one
“+” is a _____ operator.

- Unary
- **Binary**
- Ternary
- None of the above



1. Addition of new items in stack make the pointer ----- by 2 :-
 - a. Increment, bits
 - b. Increment, bytes
 - c. Decrement, bits
 - d. Decrement, bytes**
2. Next item in a linked list is known as:-
 - a. Index
 - b. Item
 - c. Node**
 - d. Child
3. What will be the postfix notation of $5+6/2$.
 - a. $56+/2$
 - b. $562+/-$
 - c. $562/+$**
 - d. $5+62/-$
4. In an AVL tree to delete a parent with two childs in a straight line following rotations will be required:-
 - a. Single
 - b. Double**
 - c. Triple
 - d. None.
5. To check the depth of an AVL tree following time will be taken:-
 - a. $1.66 \log_2 n$
 - b. $1.44 \log_2 n$**
 - c. $\log_2(n+1)-1$
 - d. $1.66 \log_2(n+1)$
6. BST is a _____ Structure:-
 - a. Linear
 - b. Non Linear**
 - c. Circular
 - d. None of Above
7. After creation of an array:-
 - a. Size can be increase but can not be decreased.
 - b. Size can be decreased but can not be increased.
 - c. Size can neither be increased nor be decreased.**
 - d. Size can be increased and can also be decreased.
8. Each node in a BST has _____ Pointers:-
 - a. 1
 - b. 2**
 - c. 3
 - d. 4
9. Highest Operators Precedence is of the following operator:-
 - a. Plus
 - b. Minus
 - c. Multiply
 - d. Exponentiation**
10. Following are the linear data structures:-
 - a. Stacks
 - b. Queues

c. Both a & b

d. None of the above

11. Each entry which points to a null value in a Singly Linked List is known as:-

a. Node

b. First Node

c. Last Node

d. Head Node

12. Non recursive calls are faster than the Recursive calls.

a. True

b. False

13. Tree data structure is a _____

a. Linear

b. Non Linear

c. Circular

d. None of Above

14. What will be the valid postfix notation of $A+B*C-D$

a. $ABC+*D-$

b. $ABC*+D-$

c. $ABCD+-*$

d. $AB+D*C$

15. When an operator is used in between two operands this is which type of notation

a. Prefix

b. Postfix

c. Infix

d. None of the Above

Question No: 1 (Marks: 1) - Please choose one

Which one of the following operations returns top value of the stack?

Push

Pop

Top

First

Question No: 2 (Marks: 1) - Please choose one

In a complete binary tree of depth 4 the number of non-leaf nodes is

7

8

15

16

Question No: 3 (Marks: 1) - Please choose one

Which of the following is NOT a linear data structure?

Linked List

Stack

Queue

Tree

Question No: 4 (Marks: 1) - Please choose one

A Linear Data Structure is the data structure in which data elements are arranged in a sequence or a linear list. Which of the following is Non Linear Data Structure?

Arrays

LinkLists

Binary Search Trees

None of these

Question No: 5 (Marks: 1) - Please choose one

In sequential access data structure, accessing any element in the data structure takes different amount of time. Tell which one of the following is sequential access data structure,

Arrays

Lists

Both of these

None of these

Question No: 6 (Marks: 1) - Please choose one

If you know the size of the data structure in advance, i.e., at compile time, which one of the following is a good data structure to use.

Array

List

Both of these

None of these

Question No: 7 (Marks: 1) - Please choose one

Recursive function calls are implemented internally using a data structure

Stack

Link-List

Tree

Queue

Question No: 8 (Marks: 1) - Please choose one

Given a stack of n items, how many POP and PUSH operations need to be performed to remove the item at its bottom?

0 POP operation and 0 PUSH operation

1 POP operation and 1 PUSH operation

n POP operations and n PUSH operations

n POP operations and n-1 PUSH operations

Question No: 9 (Marks: 1) - Please choose one

One difference between a queue and a stack is:

Queues require dynamic memory, but stacks do not.

Stacks require dynamic memory, but queues do not.

Queues use two ends of the structure; stacks use only one.

Stacks use two ends of the structure, queues use only one.

Question No: 10 (Marks: 1) - Please choose one

In the following C++ code, how many function calls are made?

```
int x, y, z;
```

```
x = 2;
```

```
y = 3 + x;
```

```
z = foobar(x,y);
```

1

4

7

8

Question No: 11 (Marks: 1) - Please choose one

Consider the following function:

```
void test_a(int n)
```

```
{
```

```
cout << n << " ";
```

```
if (n>0)
```

```
test_a(n-2);
```

}

What is printed by the call test_a(4)?

4 2
0 2 4
0 2
2 4

Question No: 12 (Marks: 1) - Please choose one

Consider the following tree,

How many leaves does it have?

2
4
6
9

Question No: 13 (Marks: 1) - Please choose one

We access elements in AVL Tree in,

Linear way only

Non Linear way only

Both linear and non linear ways

None of the given options.

Question No: 14 (Marks: 1) - Please choose one

Consider the following statements.

(i) A binary tree can contain at least 2 L Nodes at level L.

(ii) A complete binary tree of depth d is a binary tree that contains 2 L Nodes at each level L between 0 and d, both inclusive.

(iii) The total number of nodes (T_n) in a complete binary tree of depth d is $2^{d+1} - 1$.

(iv) The height of the complete binary tree can be written as $h = \log_2(T_n + 1) - 1$ where T_n is Total number of Nodes.

Which one of the following is correct in respect of the above statements regarding the Binary trees?

(i) and (iii) only

(i), (ii) and (iii) only

(ii) and (iii) only

(ii), (iii) and (iv) only

Question No: 15 (Marks: 1) - Please choose one

The following is a segment of a C program.

```
int pqr(BinaryNode t)
```

```
{ if (t == null )
```

```
return -1;
```

```
else
```

```
return 1+max(pqr(t.left),pqr(t.right)) }
```

Identify, what the above program intend(s) to do?

Compute the height of a binary tree using an in-order traversal

Compute the height of a binary tree using a pre-order traversal

Compute the depth of a binary tree using a pre-order traversal

Compute the depth of a binary tree using a post-order traversal

Question No: 16 (Marks: 1) - Please choose one

Consider the following infix expression.

$3 * 2 * 5 - 1$

If one converts the above expression into postfix, what would be the resultant expression?

$3 * 2 * 5 * 1 -$

$3 * 2 * 5 * 1 -$

$3 * 2 * 5 - 1$

3 2 5 * 1 -

